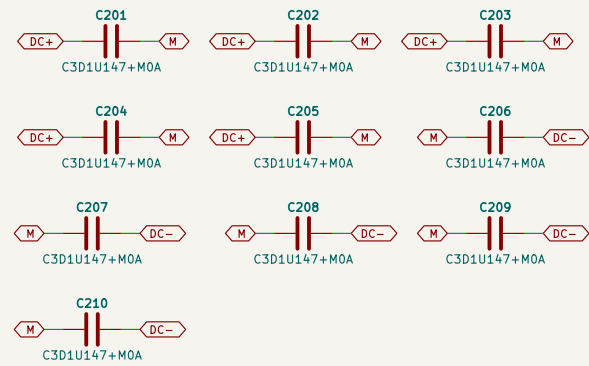
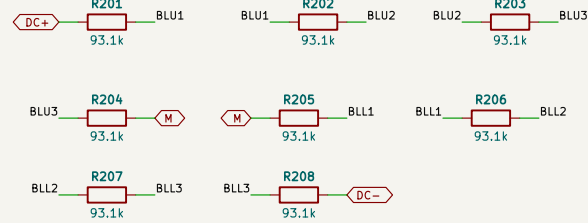


SPLIT DC LINK



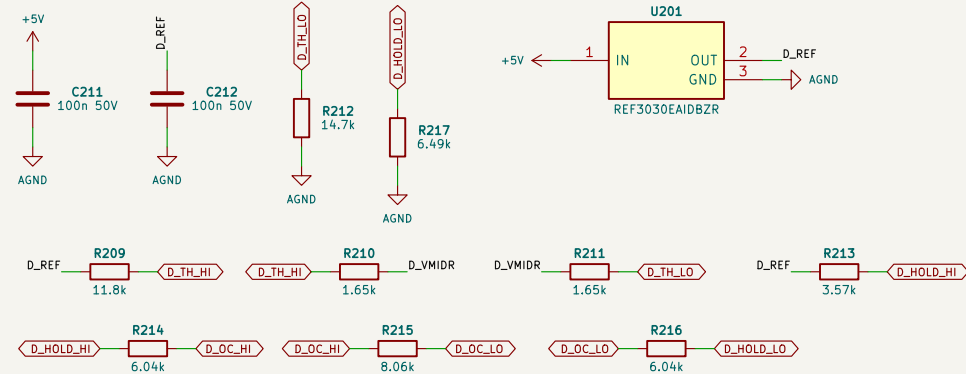
Upper half DC+ to M, lower half M to DC-, 5 x 140 uF each (700 uF per half, 350 uF in series).
M is the C_f star point of the LCL filter (no midpoint control: two-level).

BLEEDERS (BALANCE THE HALVES, DISCHARGE THE LINK)



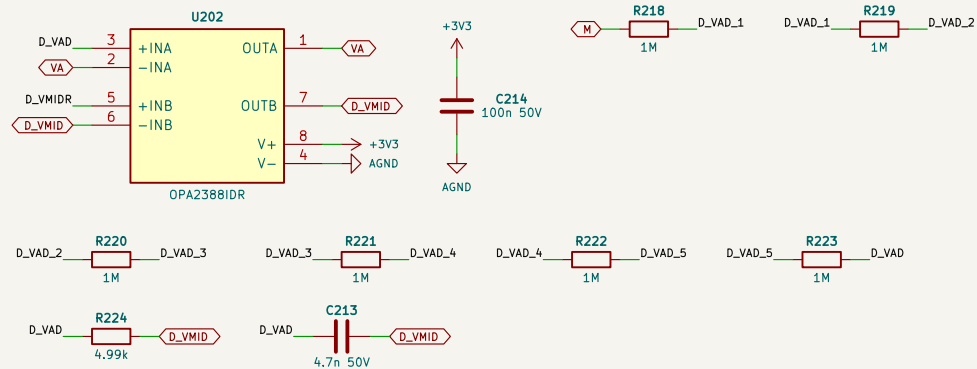
4 x 93.1k 1210 per half (372.4k); label 'wait 15 min'.

REFERENCE (PORT.LEAN_REFS: REF3030E, VMID 1.646 V AND THE DC-PORT THRESHOLD LADDERS)



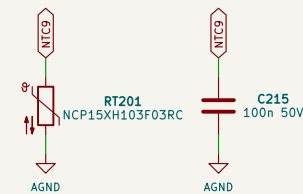
Shared with the DC port (stage 4). VMID buffered here; IA, IA_H, VAX of PCS_PC sit on it.

MIDPOINT DIVIDER (6 X 1 M ARHV06 + 4.99 K TO VMID, 4.7 NF: 1 V = 1203 V)



VA = VMID + V(M)/1203 (OPA2388 follower); its second half is the VMID follower. VB (DC+) comes from the DC port's bank divider (lean_dividers).

DC-LINK TEMPERATURE (NTC9 TO PCS_X)



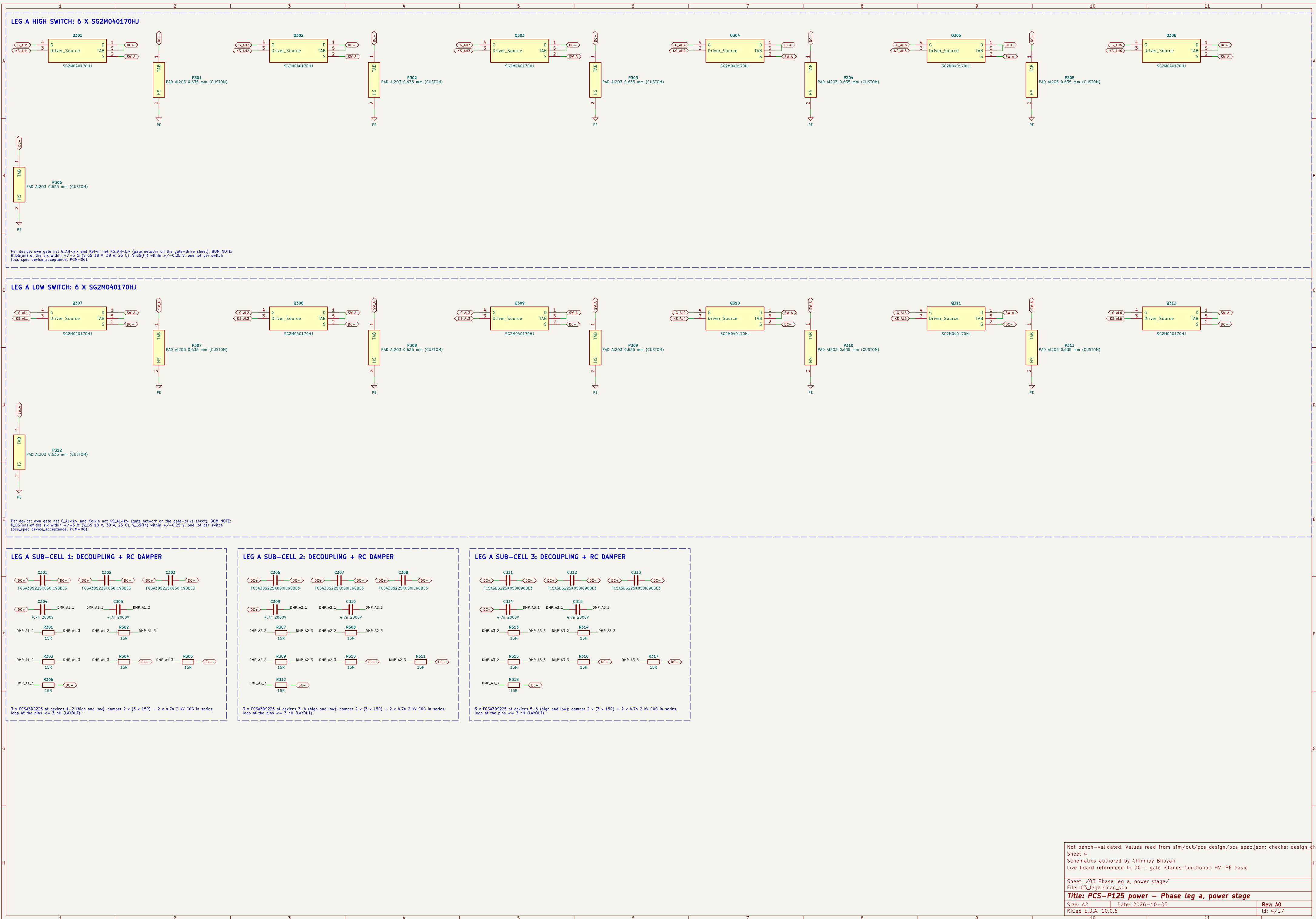
Murata NCP15 10 k on the board between the film capacitors; biased on the control board.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 3
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

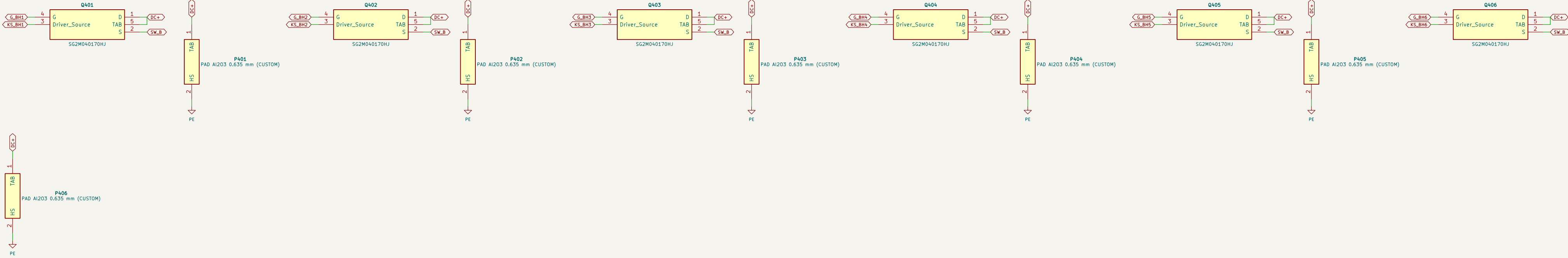
Sheet: /02 DC link, bleeders, sensing/
File: 02_dclink.kicad_sch

Title: PCS-P125 power – DC link, bleeders, sensing

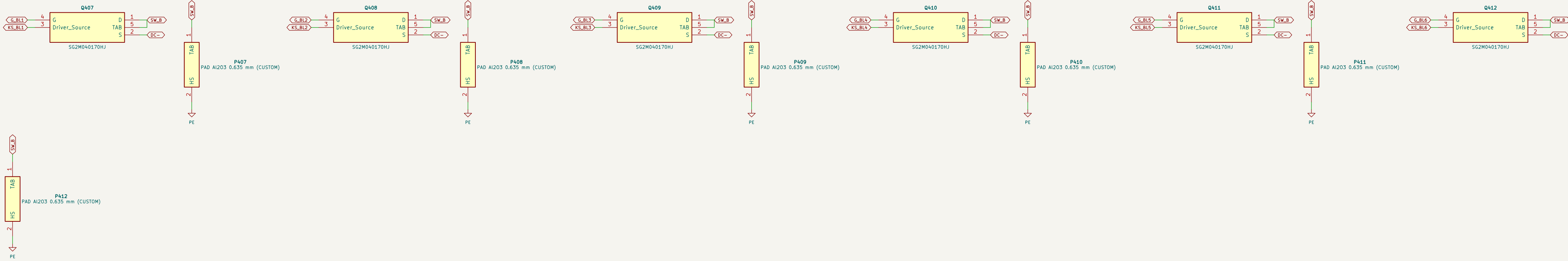
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 3/27



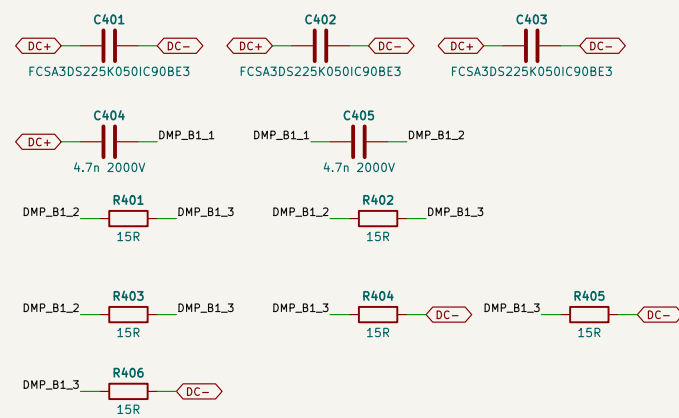
LEG B HIGH SWITCH: 6 X SG2M040170HJ



LEG B LOW SWITCH: 6 X SG2M040170HJ

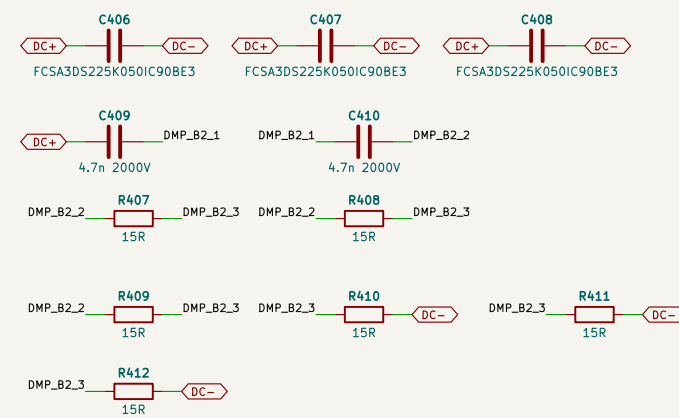


LEG B SUB-CELL 1: DECOUPLING + RC DAMPER



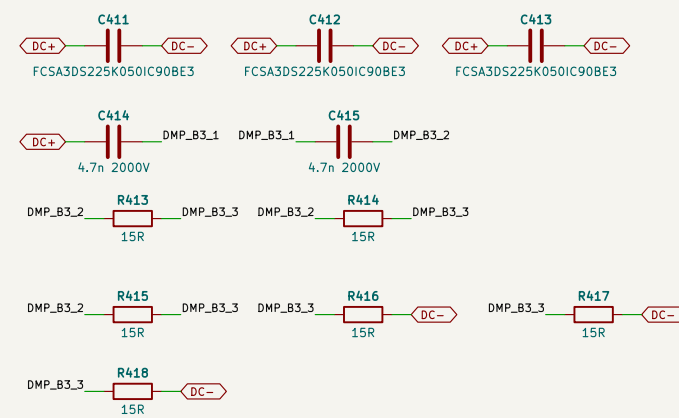
3 x FCSA3DS225 at devices 1-2 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG B SUB-CELL 2: DECOUPLING + RC DAMPER



3 x FCSA3DS225 at devices 3-4 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

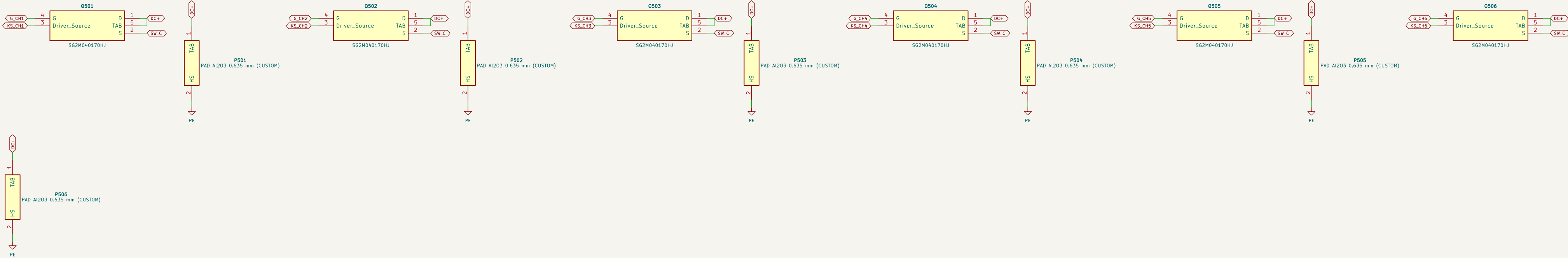
LEG B SUB-CELL 3: DECOUPLING + RC DAMPER



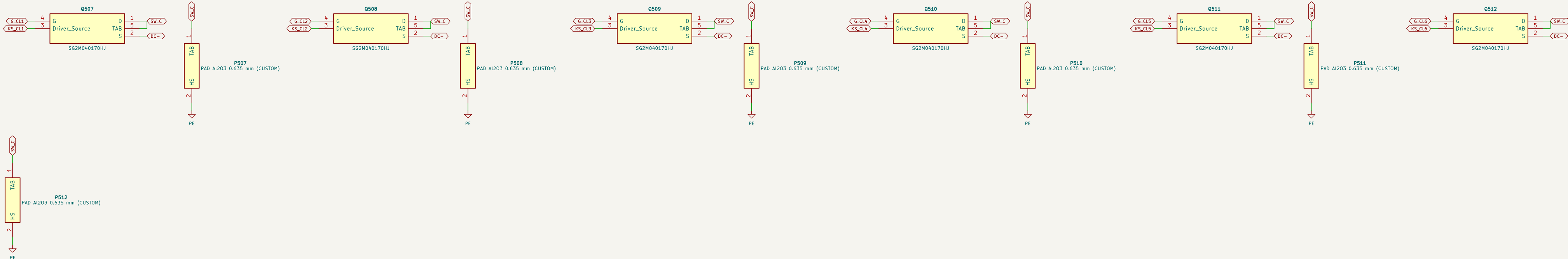
3 x FCSA3DS225 at devices 5-6 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 5
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

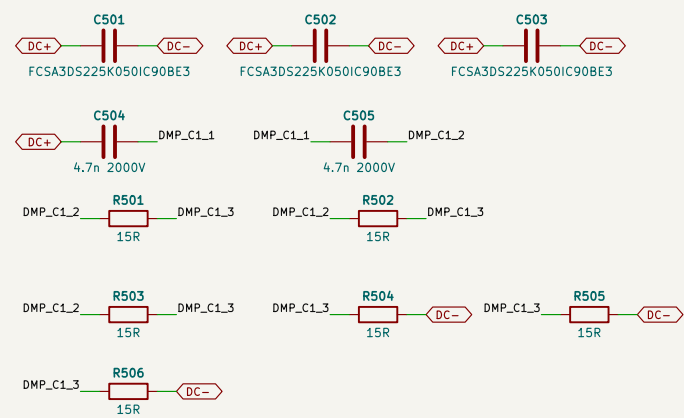
LEG C HIGH SWITCH: 6 X SG2M040170HJ



LEG C LOW SWITCH: 6 X SG2M040170HJ

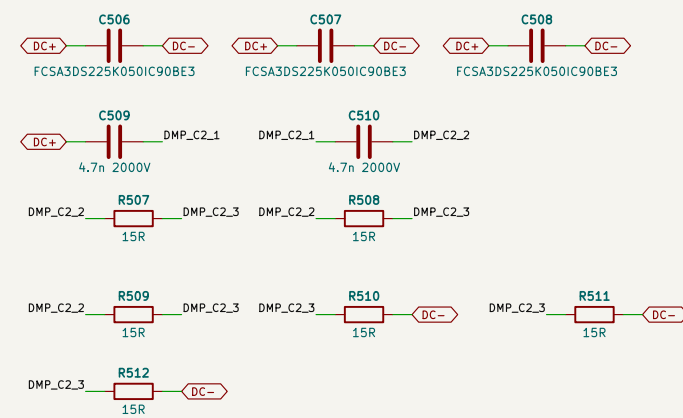


LEG C SUB-CELL 1: DECOUPLING + RC DAMPER



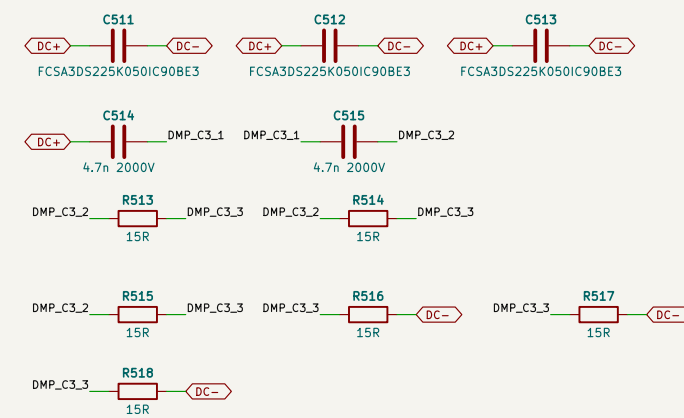
3 x FC5A30S225 at devices 1-2 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG C SUB-CELL 2: DECOUPLING + RC DAMPER



3 x FC5A30S225 at devices 3-4 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

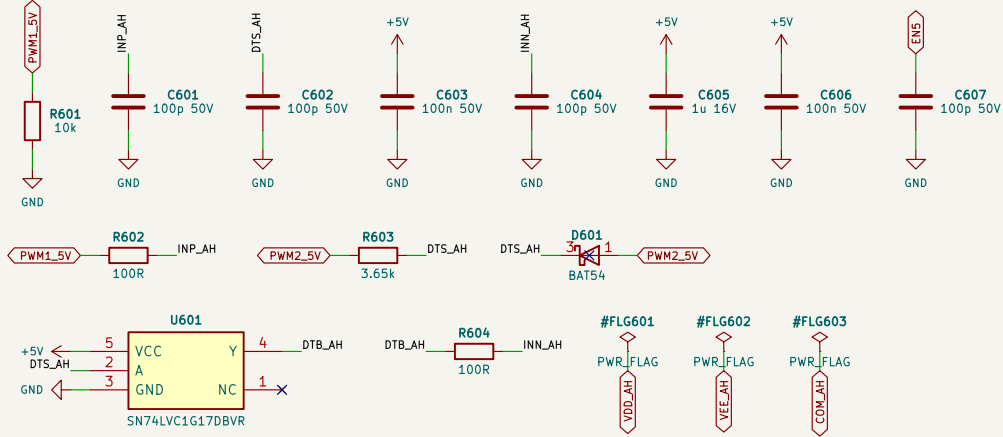
LEG C SUB-CELL 3: DECOUPLING + RC DAMPER



3 x FC5A30S225 at devices 5-6 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

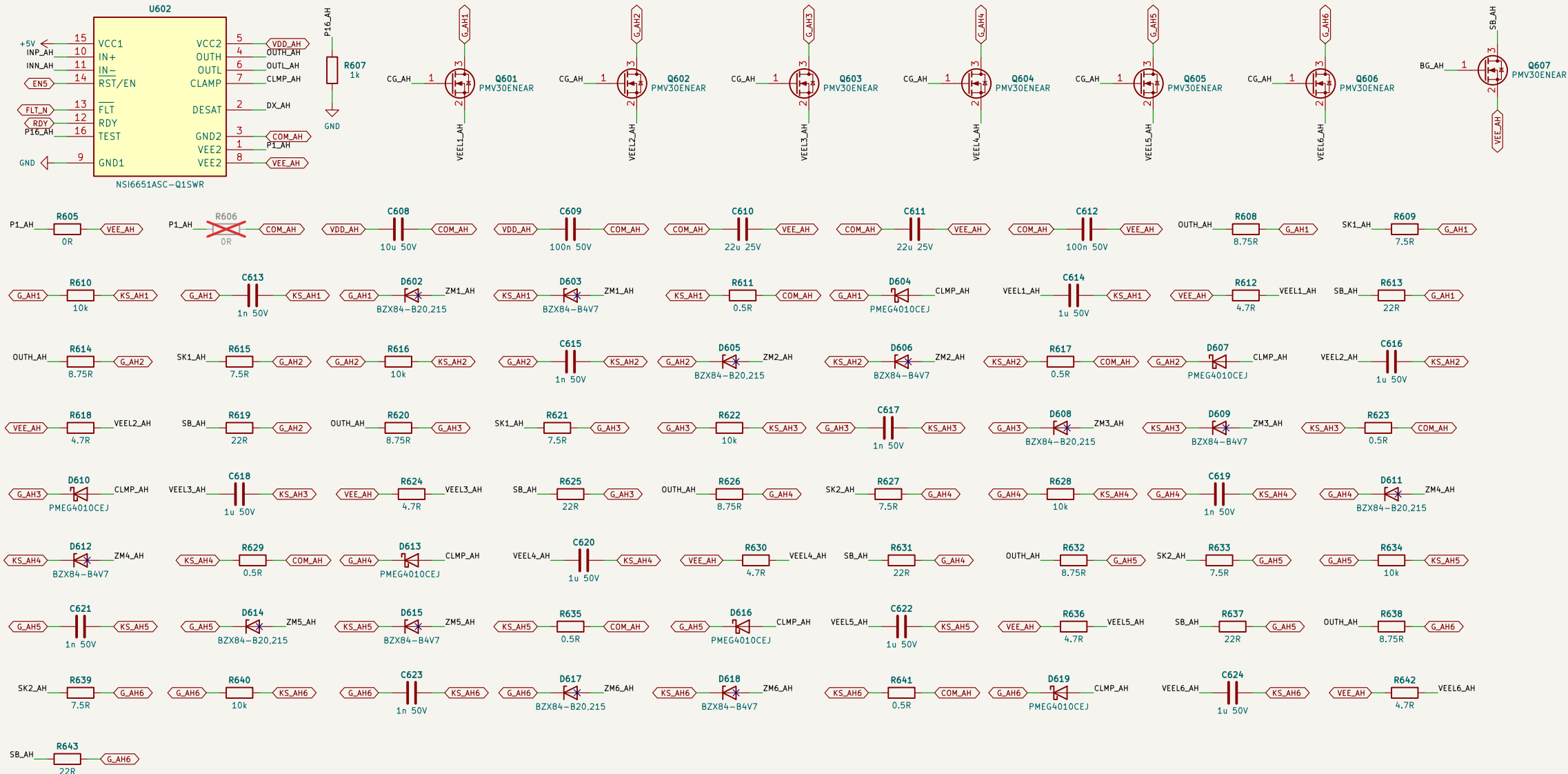
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 6
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

AH: LOGIC SIDE



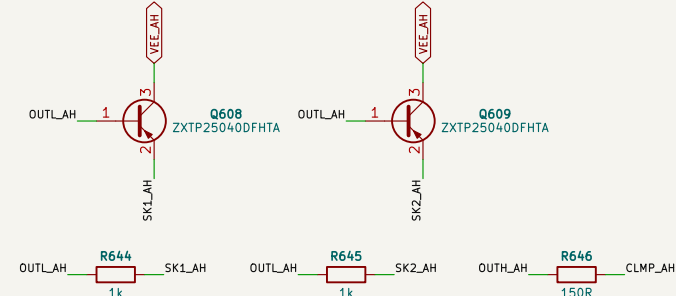
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

AH: DRIVER AND 6 GATES



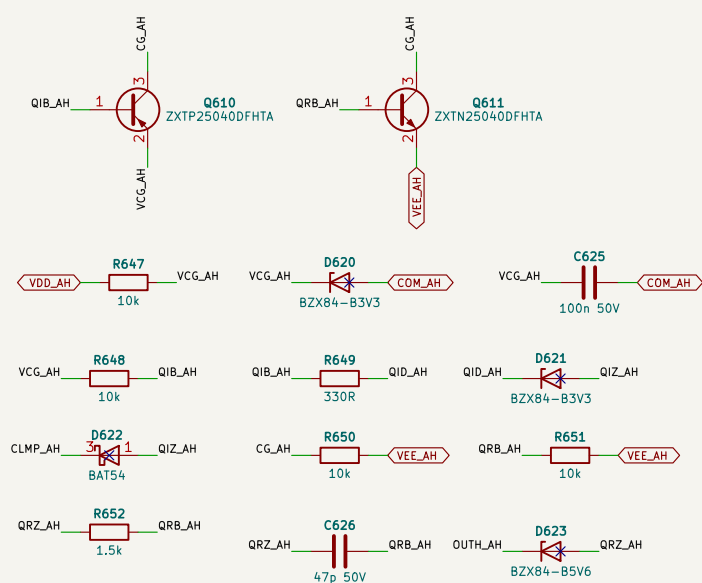
Per device: Ron/Reff. 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

AH: TURN-OFF BUFFER



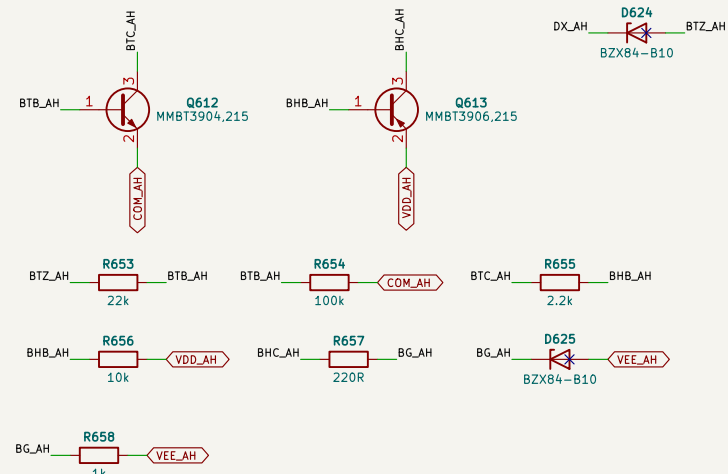
2 x ZXTP25040DFHT emitter followers on OUTL (B-E 1k, C to VEE): each sinks the Roff current of 3 gates; the NSI6651 supplies base current only. Turn-on stays on OUTH.

AH: MILLER CLAMP CONTROL



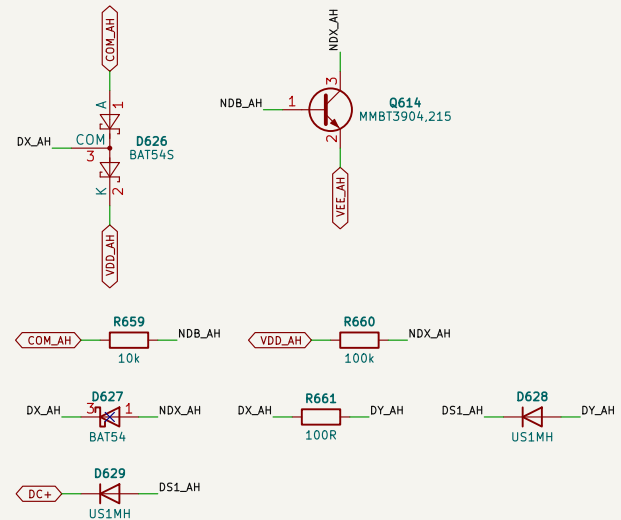
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

AH: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V, per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

AH: DESAT



DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs ~ 2 x US1M - drain. BAT545 keeps DX against COM and VDD against dv/dt through the string; Cbik footprint DNP.

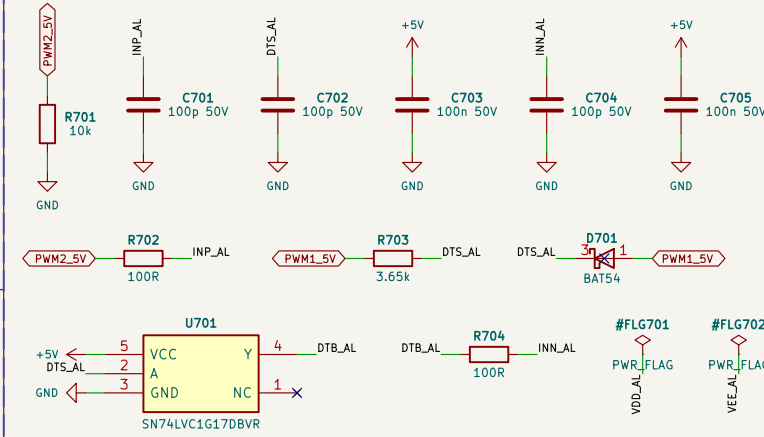
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 7
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /06 Leg a high-side gate drive/
File: 06_gdah.kicad_sch

Title: PCS-P125 power - Leg a high-side gate drive

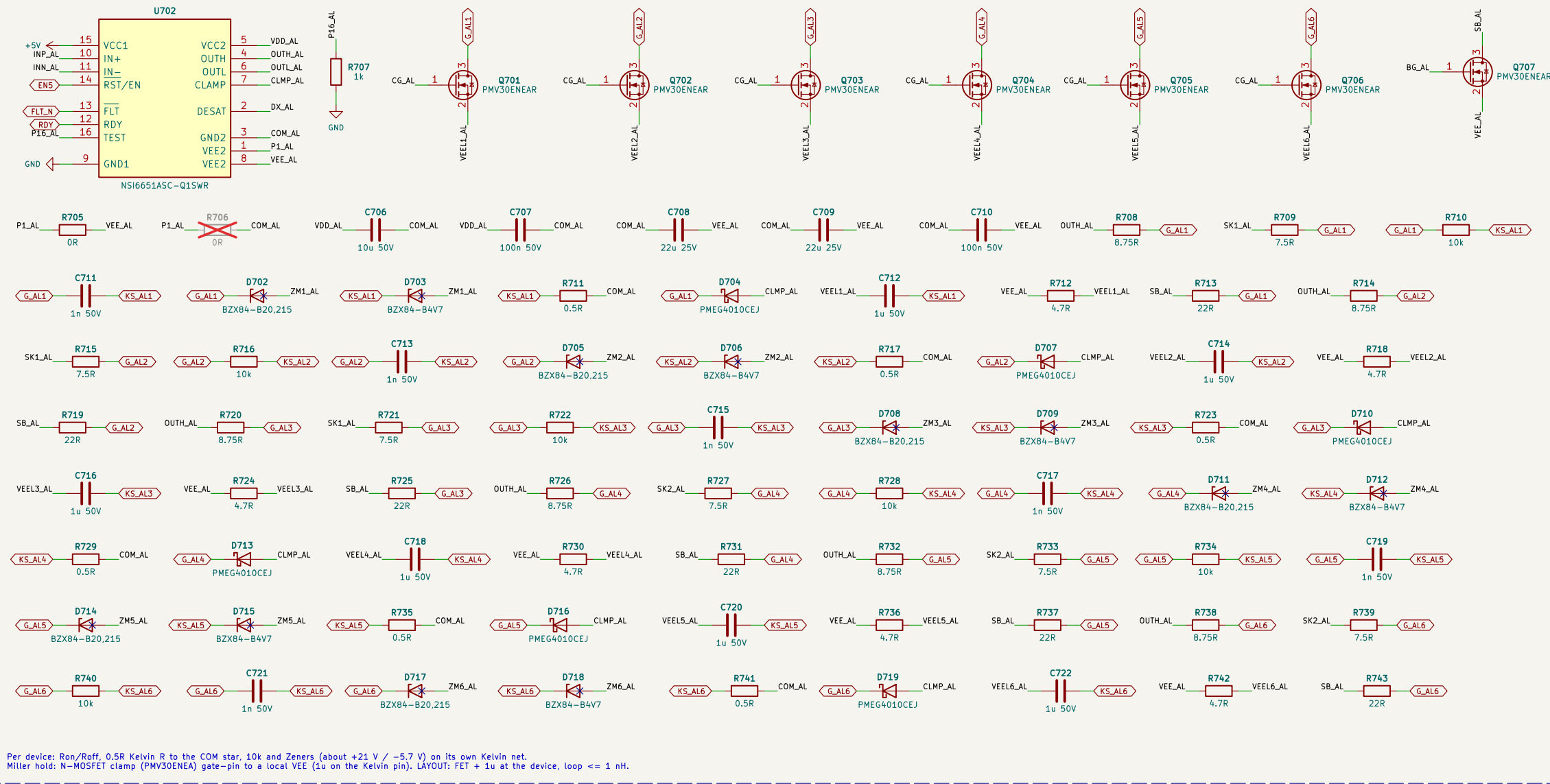
Size: A2 Date: 2026-10-05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 7/27

AL: LOGIC SIDE



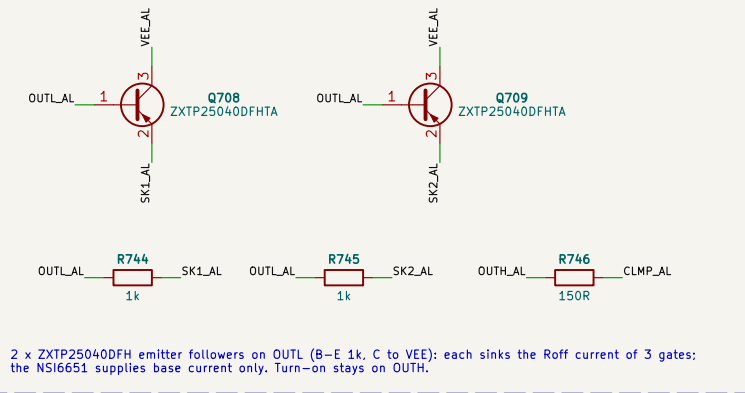
IN+ = own PWM; IN- = complementary switch, falling edge detected 3.6V/100p.
 (C601C6 = 10k dead time = deadTime; hardware); Data PS (active low) pulls RDP low.

AL: DRIVER AND 6 GATES



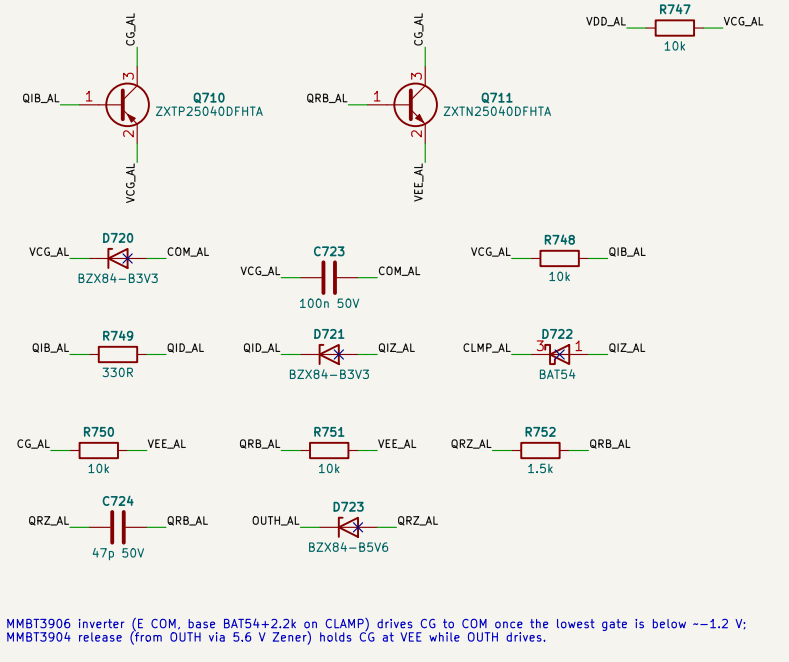
Per device: Rsw/Rdsw: 0.58 Kelvin @ to the COM star, 10k and 20k (about +21 V / -5.7 V) on its own Kelvin net.
 Miller mode to MOSFET clamp (PHE04010CE) gate-pin to a base VEE (to on the Kelvin pin) LATCH: ET = 1u at the device, loop <= 1 ns.

AL: TURN-OFF BUFFER



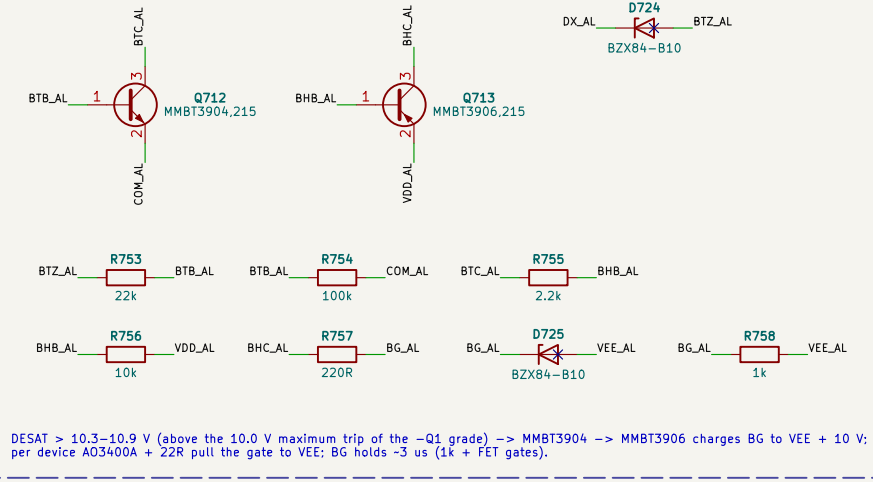
2 x 2XTP25040DPH1x another follows on OUTL (B-E is C to VEE) each sinks the Roff current of 3 gates:
 the MOSFETs supply base current only, turn-on delay on OUTL.

AL: MILLER CLAMP CONTROL



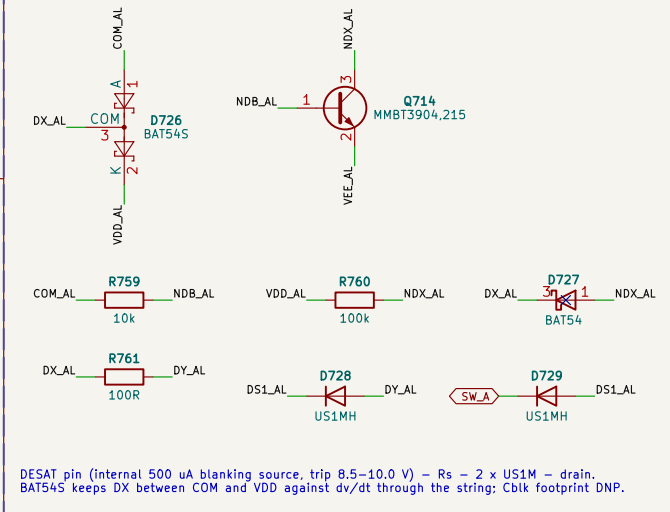
MHBT3906: Inverter (C COM, base B8204-2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
 MHBT3906: release (from OUTP via 5.6 V Zener) holds CG at VEE while OUTP driver.

AL: SHORT-CIRCUIT BOOSTER



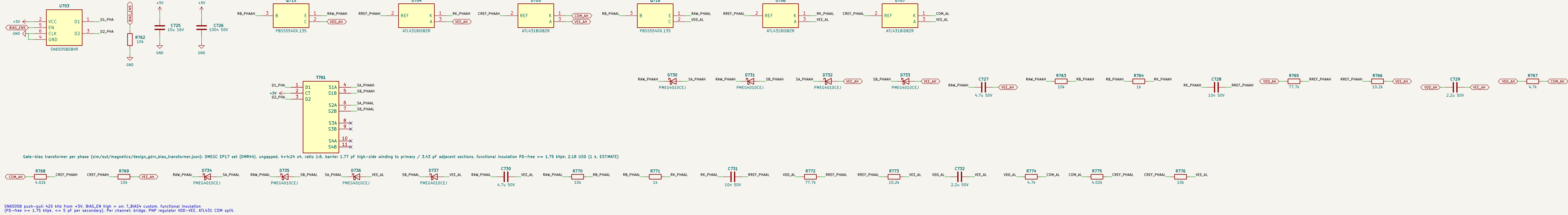
DESAT = 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) --> MHBT3904 --> MHBT3906 charges BG to VEE + 10 V
 per device A03404 = 22M pull the gate to VEE; BG holds -3 on (14 + FET gates).

AL: DESAT



DESAT pin (Interval 500 uA blanking source, trip 8.5-10.0 V) - Rg = 2 x USM - drain
 B8204 keeps BG between COM and VEE against Rv/G through the wiring; C8B Redundant ONP.

PHA: GATE BIAS (PER PHASE)

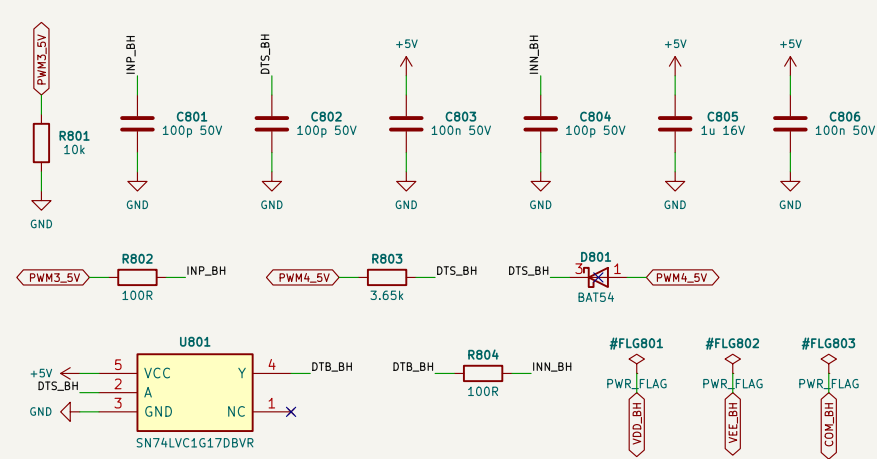


Gate-bias transformer per phase (sim/out/magnetics/design_gdnv/bias_transformer.json): DMSC EP17 set (DMR4), ungapped, 4+424 vA, ratio 1:6, barrier 1.77 pF of high-side winding to primary / 3.43 pF adjacent sections, functional insulation PD-free >= 1.75 kV@ 2.18 USD (± k, ESTIMATE)

LEG A: LAYOUT RULES

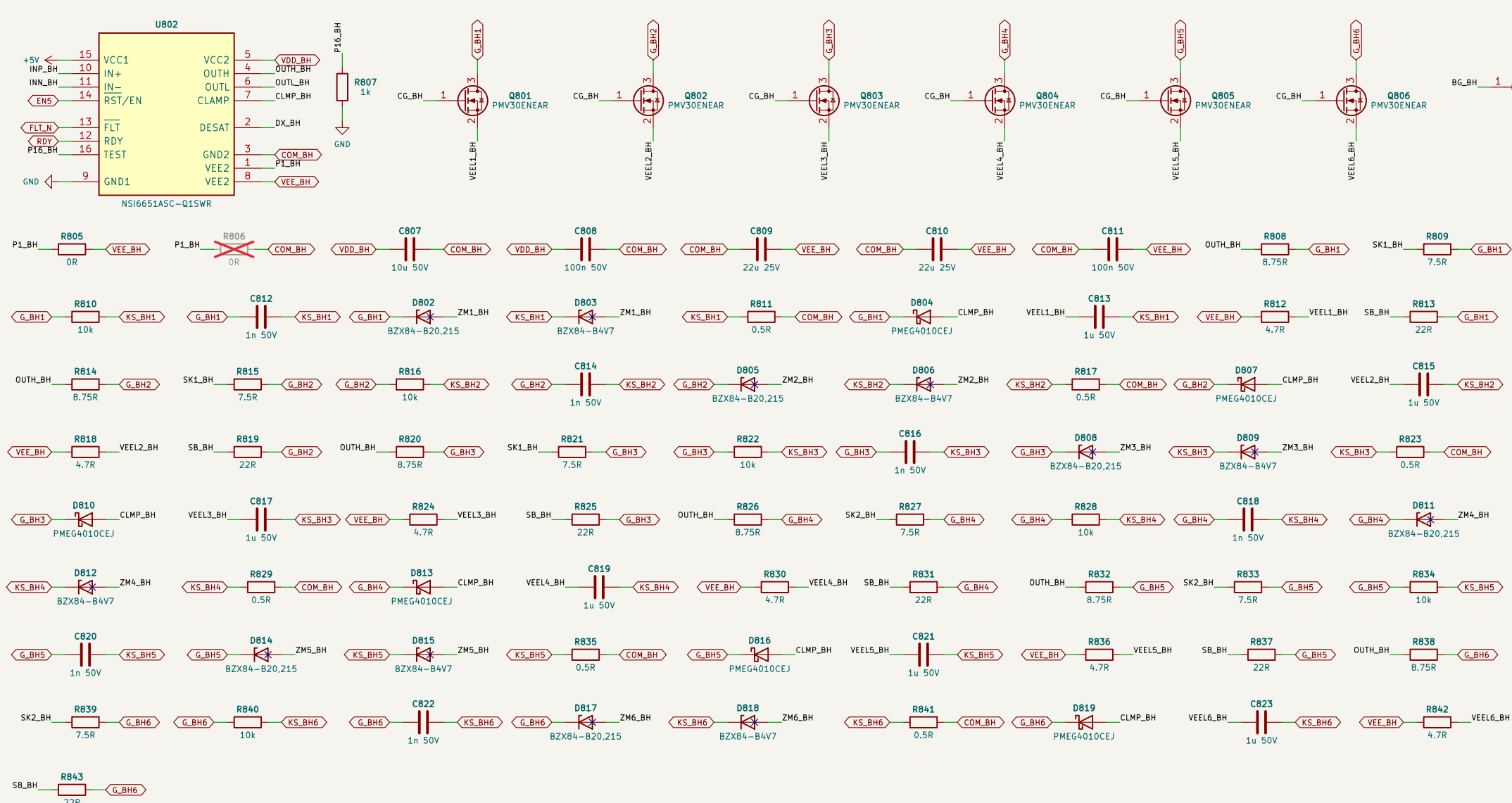
Gate and Kelvin of each device as coupled pairs of equal length from the COM star; Roff Kelvin @ 10k ± 1u
 Zener and the clamp FET = C.A. at each point (Roff = 50u), PNP follower next to their three gates; DESAT
 wiring >= 2.5 mm sag gap per block, cool to VEE; Miller hold of all gates NOT elsewhere - BENCH TEST.

BH: LOGIC SIDE



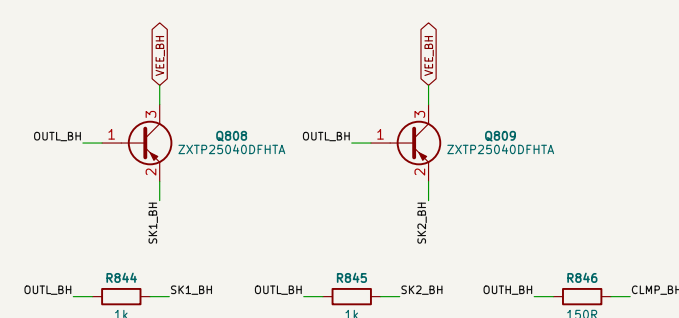
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

BH: DRIVER AND 6 GATES



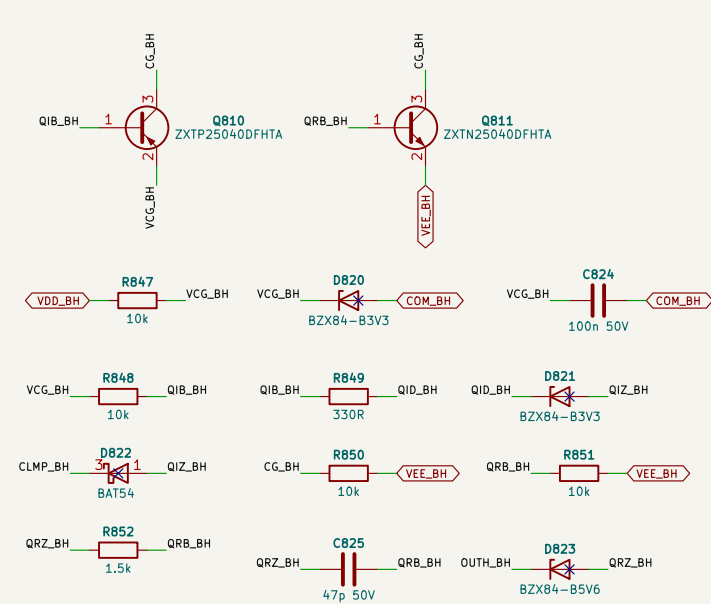
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

BH: TURN-OFF BUFFER



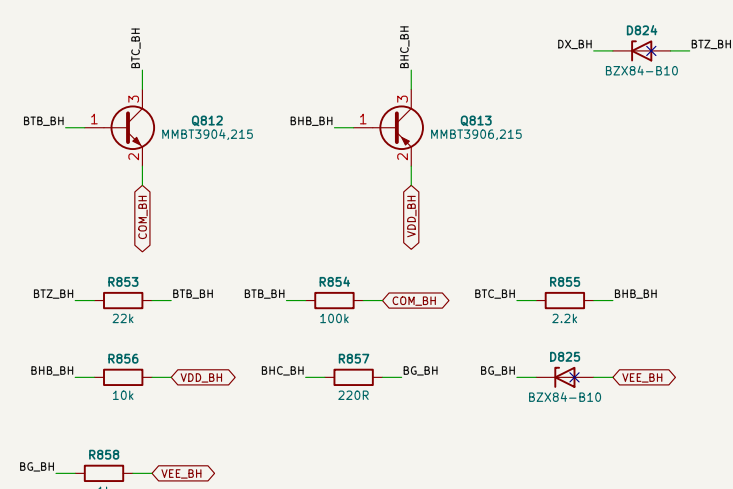
2 x XZPT25040DFH emitter followers on OUTL (B-E 1k, C to VEE): each sinks the Roff current of 3 gates; the NSI6651 supplies base current only. Turn-on stays on OUTH.

BH: MILLER CLAMP CONTROL



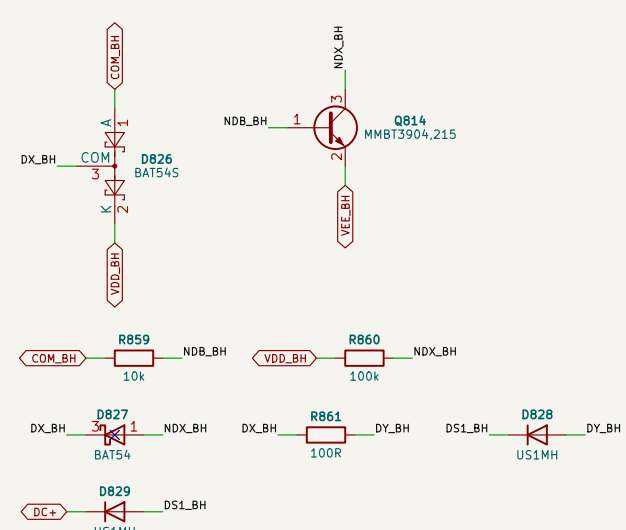
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

BH: SHORT-CIRCUIT BOOSTER



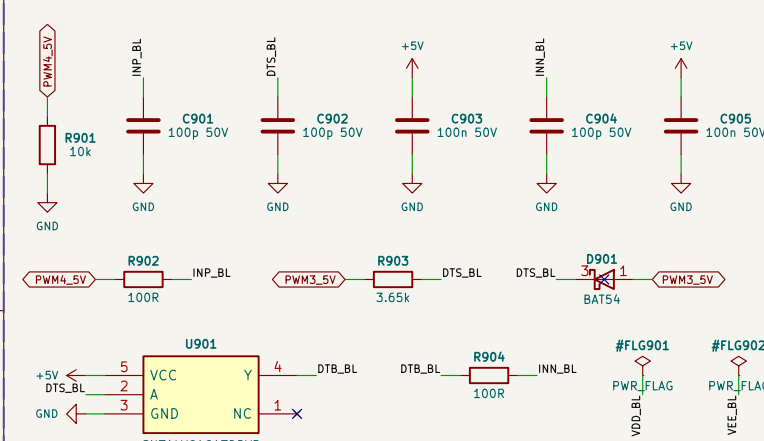
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

BH: DESAT



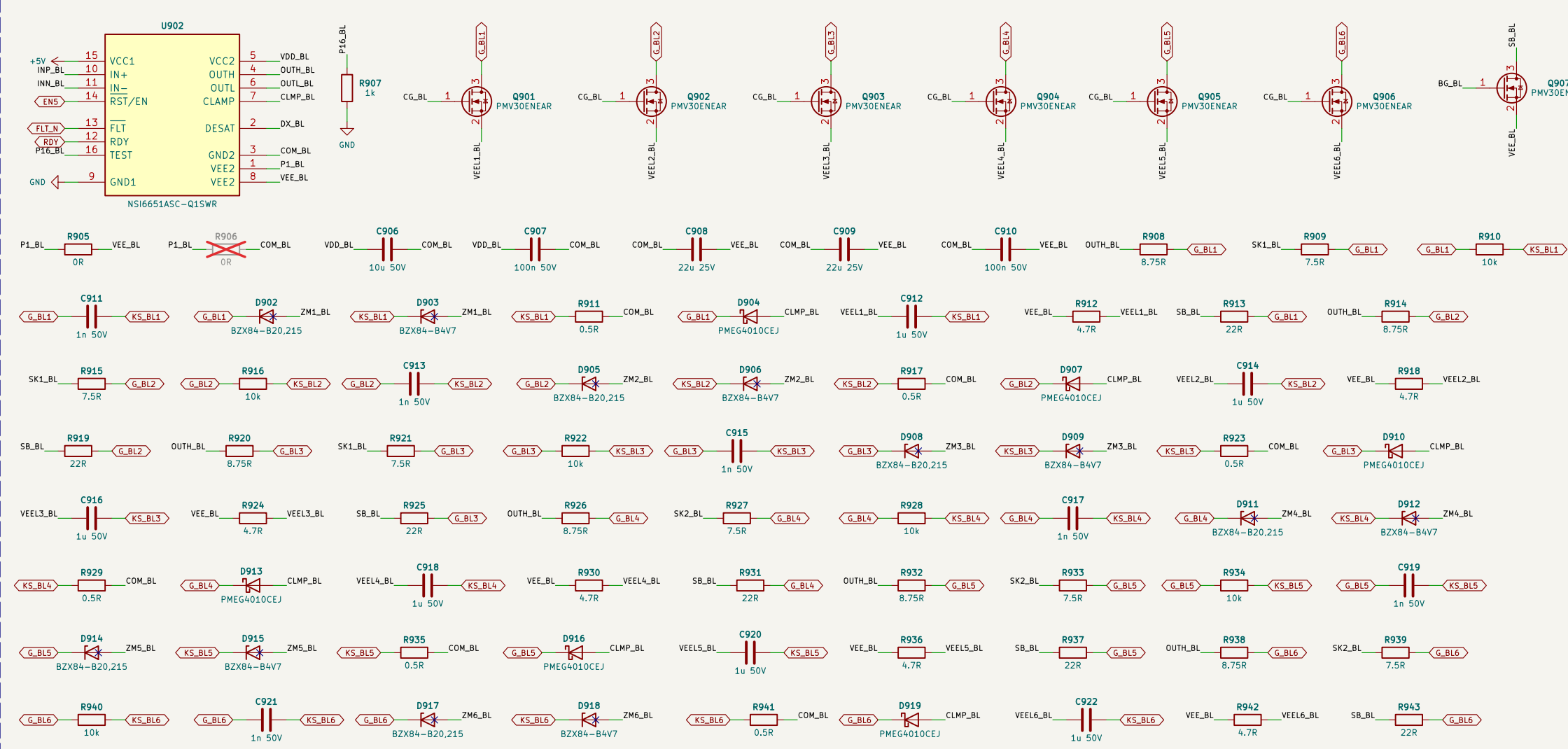
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string: Cblk footprint DNP.

BL: LOGIC SIDE



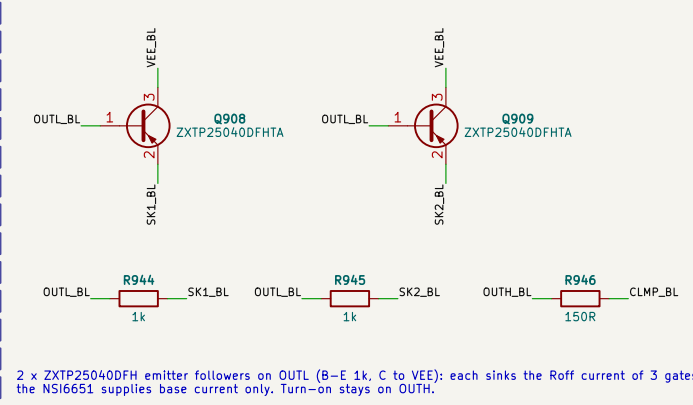
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls I

BL: DRIVER AND 6 GATES



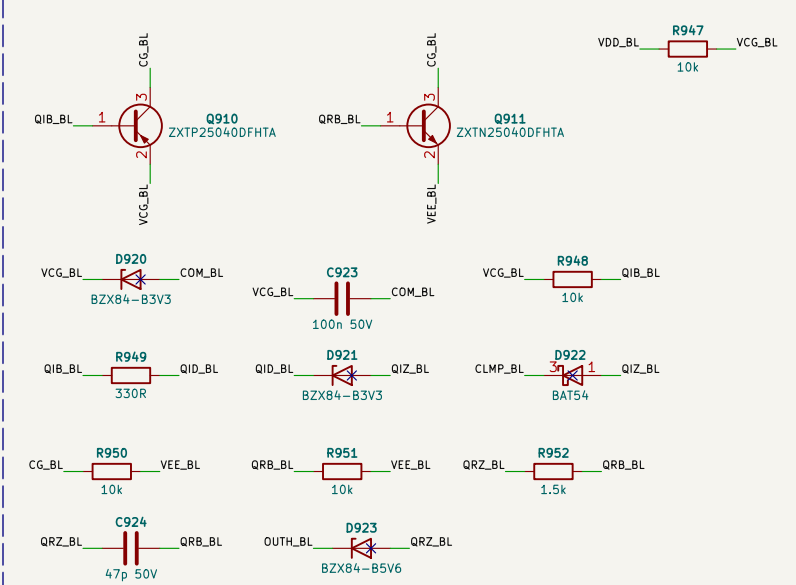
Per device: Ros/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

BL: TURN-OFF BUFFER



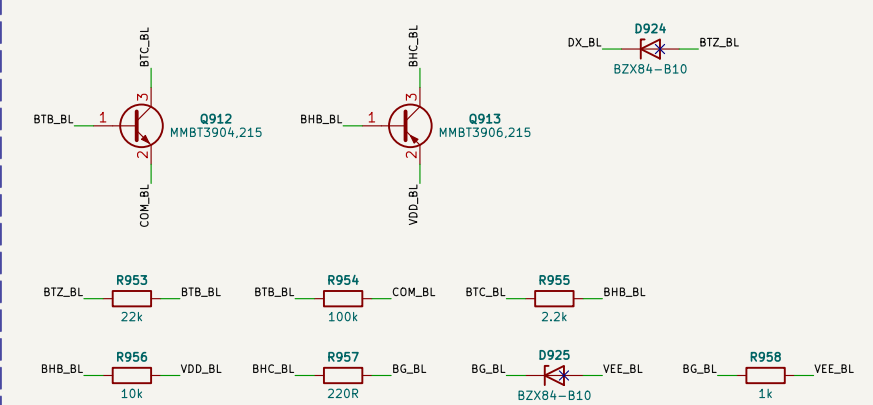
2 x ZXTP250400FH emitter followers on OUTL (B-E 1k, C to VEE); each sinks the Roff current of 3 gate the NS16651 supplies base current only. Turn-on stays on OUTH.

BL: MILLER CLAMP CONTROL



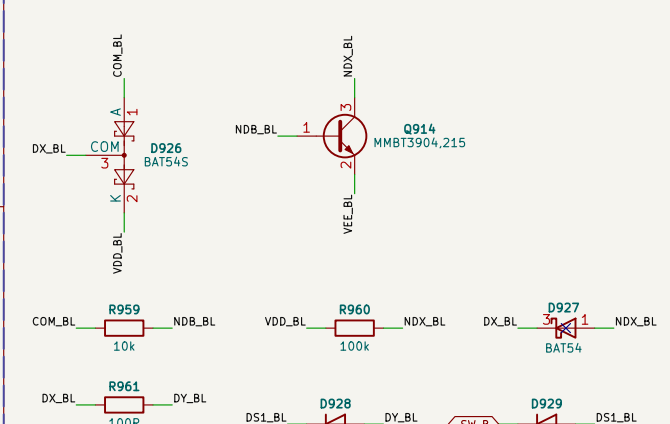
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V. MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

BL: SHORT-CIRCUIT BOOSTER



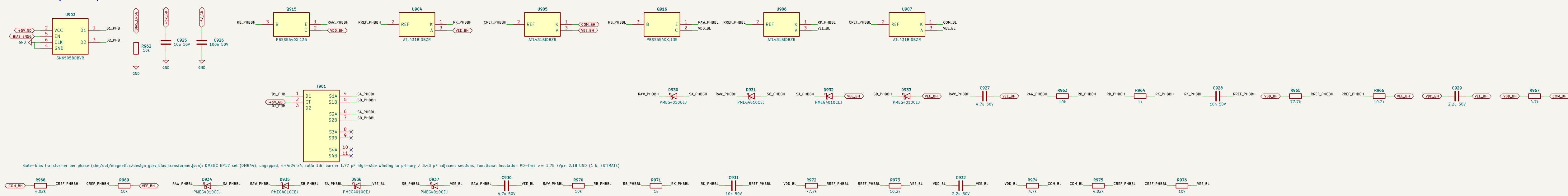
DESAT > 10.3–10.9 V (above the 10.0 V maximum trip of the -Q1 grade) → MMB per device AD3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

BL: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) = $R_s = 2 \times U_{SIH} = \text{drain}$.
 BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNF

PHB: GATE BIAS (PER PHASE)

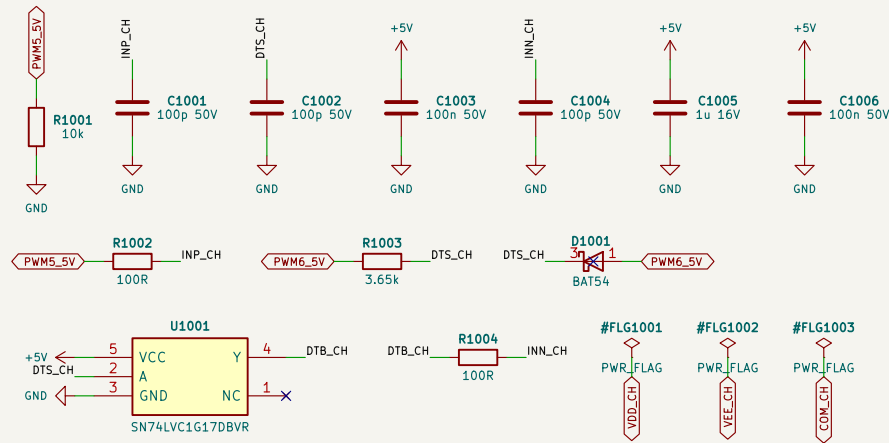


SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation (PD-free ≥ 1.75 kVp, ≤ 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

LEG B: LAYOUT RULES

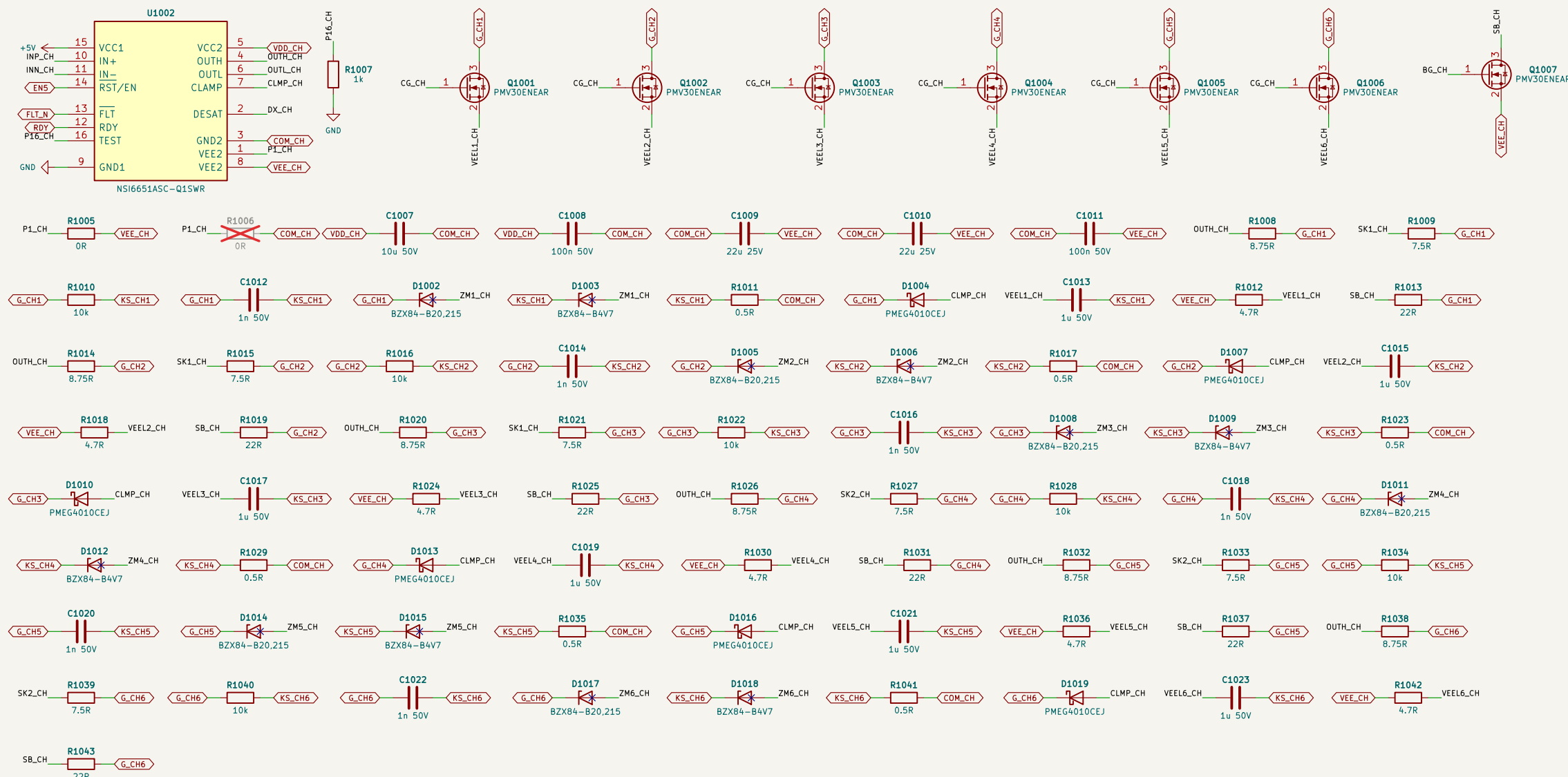
Gate and Kelvin of each device as coupled pairs of equal length from the COM star; Ron, Roff, Kelvin R, 10k + 1n Zeners and the clamp FET + C_{VL} at each device (isop. <= 1 nH). PNP followers next to their three gates. DESAT string >= 2.5 mm pad gap per diode; coat to PDI. Miller hold of six gates NOT simulated = BENCH TEST.

CH: LOGIC SIDE



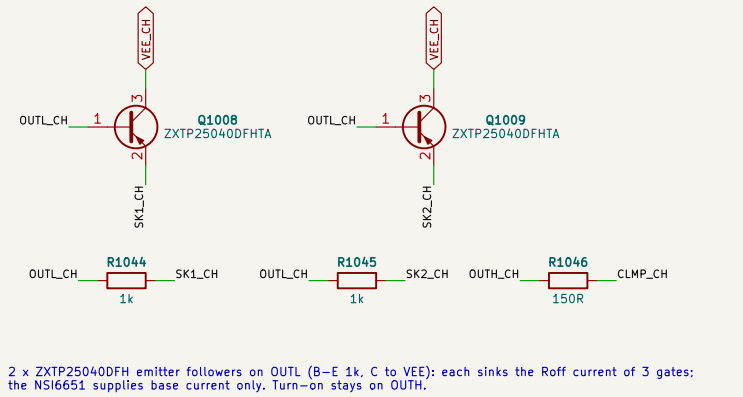
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

CH: DRIVER AND 6 GATES



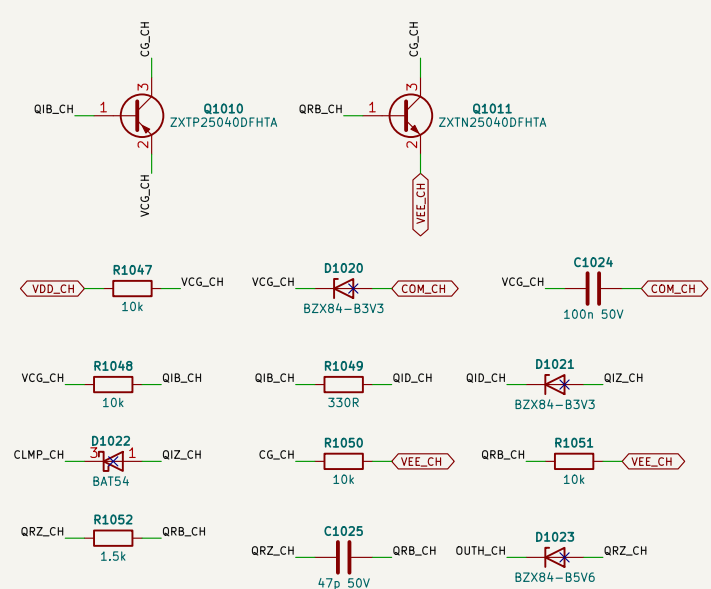
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin), LAYOUT: FET + 1u at the device, loop <= 1 nH.

CH: TURN-OFF BUFFER



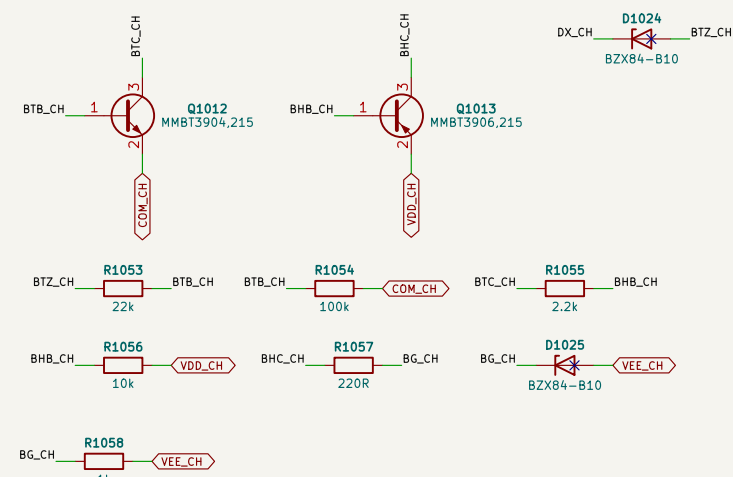
2 x ZXTP25040DFHTA emitter followers on OUTL (8-T 1k, C to VEE): each sinks the Roff current of 3 gates; the NSI6651 supplies base current only. Turn-on stays on OUTH.

CH: MILLER CLAMP CONTROL



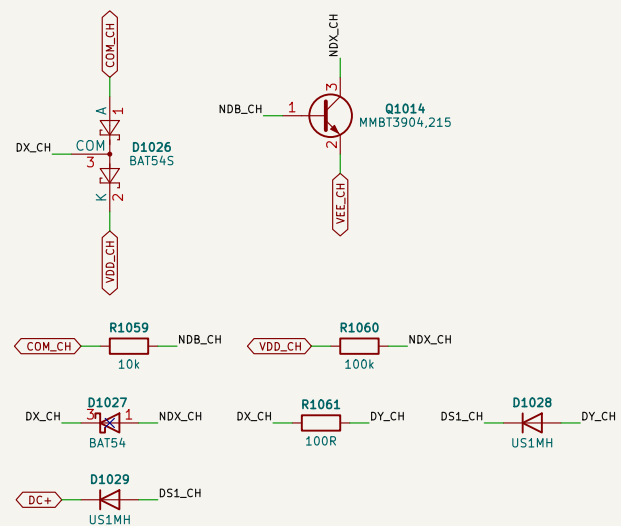
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

CH: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V, per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

CH: DESAT

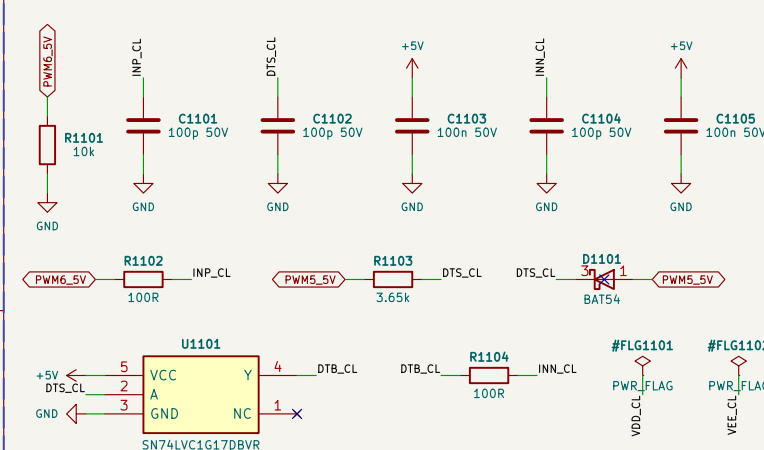


DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs ~ 2 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 11
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

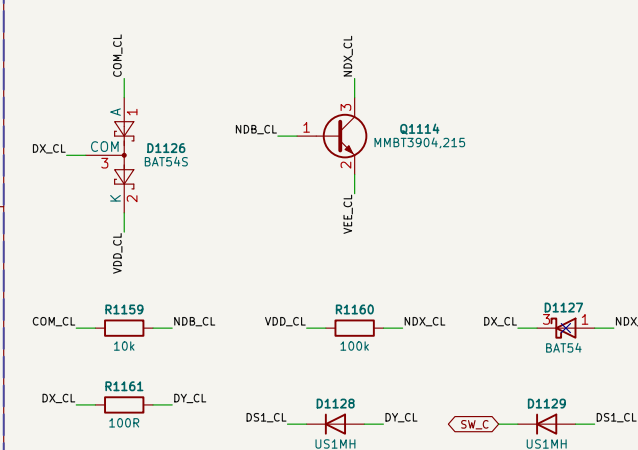
Sheet: /10 Leg c high-side gate drive/
File: 10_gdch.kicad_sch
Title: PCS-P125 power - Leg c high-side gate drive
Size: A2 Date: 2026-10-05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 11/27

CL: LOGIC SIDE



IN+ = own PWR; IN- = complementary switch, falling edge detected 3.6V/100p.
COM1CL = this dead time = deadTime; turnOnCL; Bus F0 (active low) pulls RDP low.

CL: DESAT

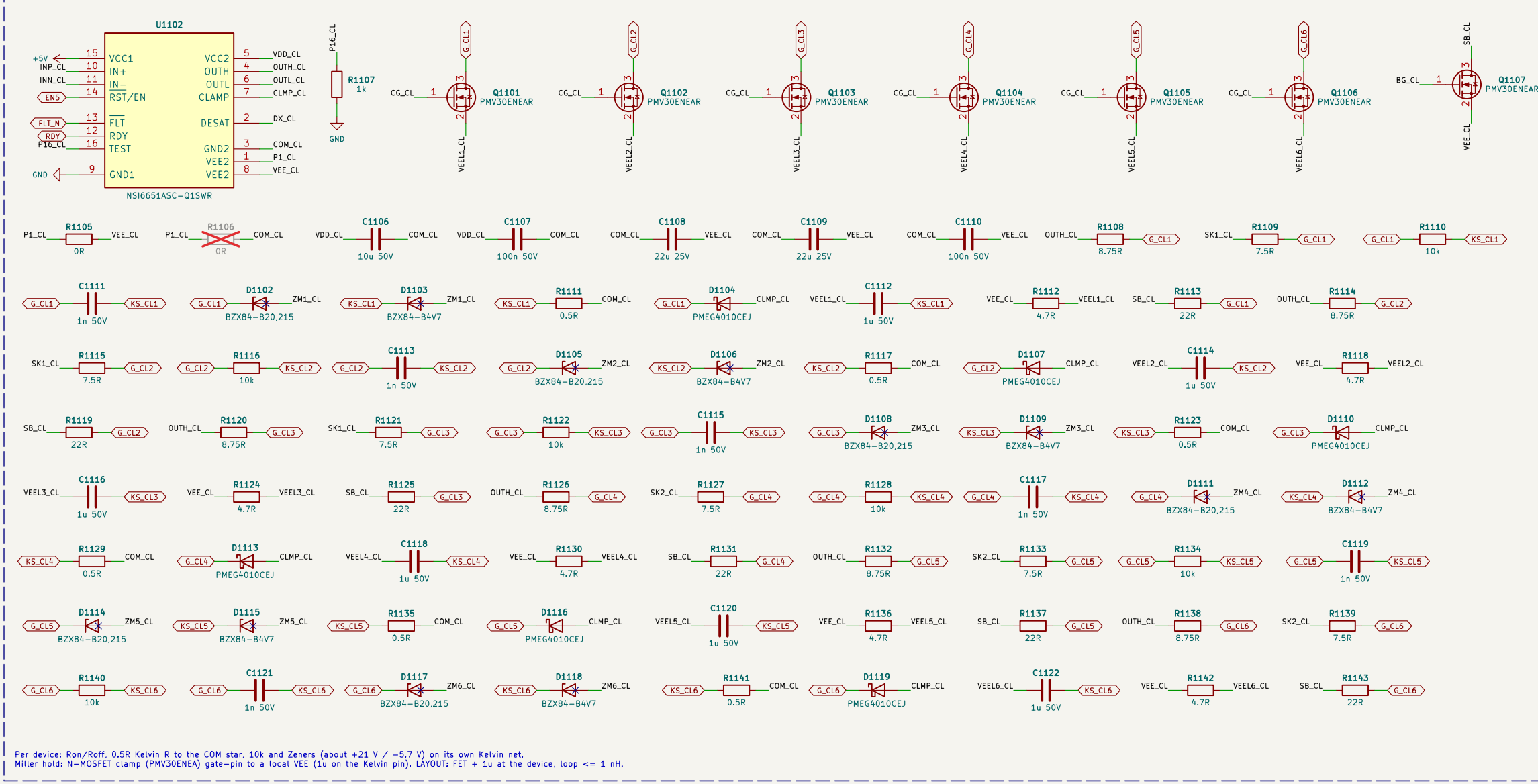


DESAT pin (interval 500 ns blanking source, trip 8.5-10.0 V) - R₁₀ = 2 x USM - drain.
BATTAG keeps 5V between COM and VDD against RvD through the wiring. CSB Redundant ONP.

LEG C: LAYOUT RULES

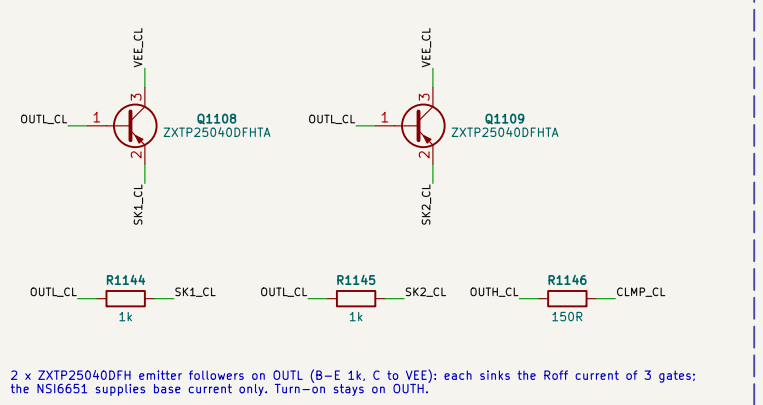
Gate and Keats of each device are coupled pairs of equal length from the COM start; Ron; Roff; Keats; R₁₀ = 1n.
Zener and the clamp FET = C₁₀ at each device (loop = 3 mil). PNP follows next to their three gates. DESAT
driving == 2.5 mm pad gap per diode; coat to PDS. Miller hold of six gates NOT simulated - BENCH TEST.

CL: DRIVER AND 6 GATES



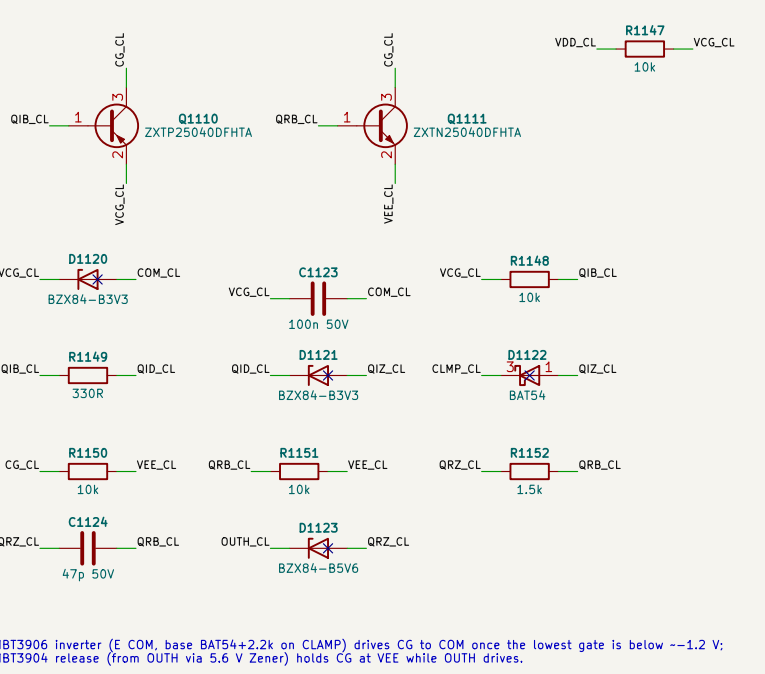
Per device: R_{on}/R_{off} 0.5R Keats 8 to the COM star. 10k and Zeners (about +21 V - -5.7 V) on its own Kelvin VEE.
Miller hold to MOSFET clamp (PNEG401DCE) gate-pin to a base VEE (to on the Kelvin pin) LATCHOUT FET = 1n at the device, loop == 1 nH.

CL: TURN-OFF BUFFER



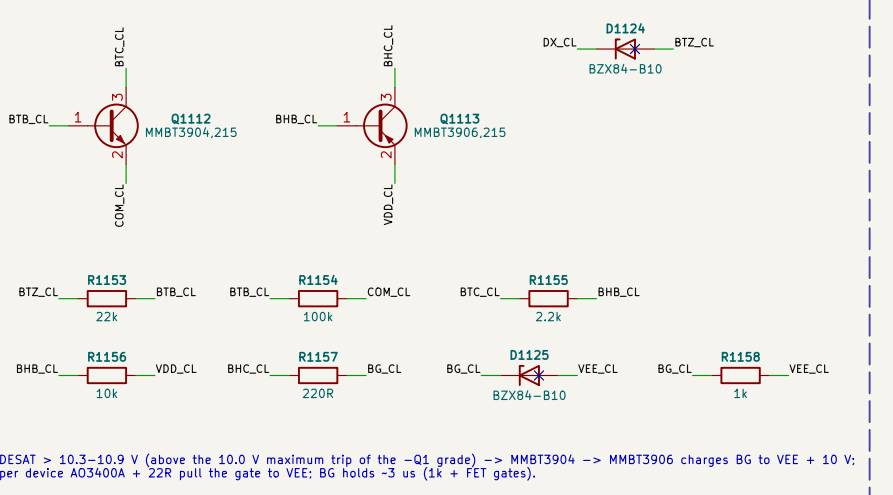
2 x ZTP25040DPHTA emitter followers on OUTL (B-E 1k, C to VEE) each sinks the Buff current of 3 gates;
the R1106/107 supplies base current only. Turn-on: R_{on} = 0.5R_{on}.

CL: MILLER CLAMP CONTROL



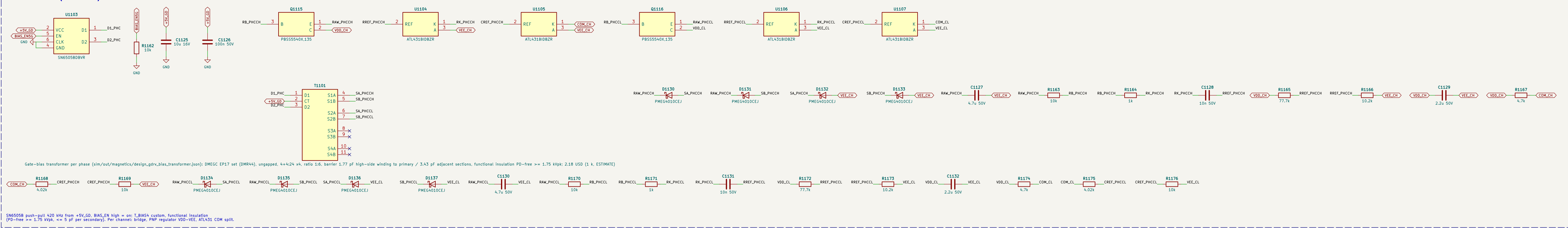
MMBT3906 (active E COM base BATTAG=2.2k on CLAMP) drives CC to COM once the lowest gate is below -1.2 V;
MMBT3904 (release (from OUTL via 5.6 V Zener) inside CC at VEE while OUTL drives).

CL: SHORT-CIRCUIT BOOSTER



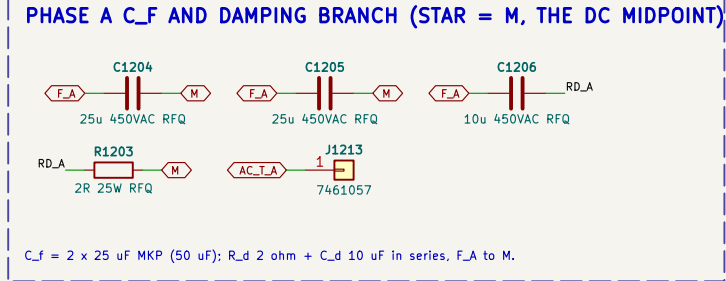
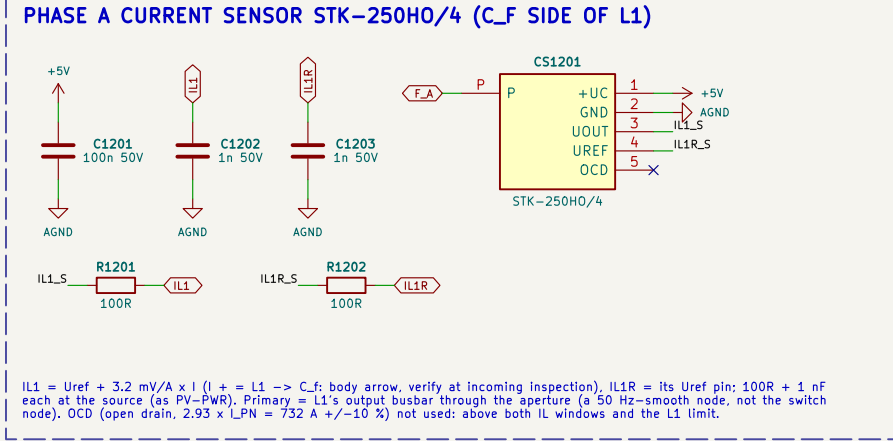
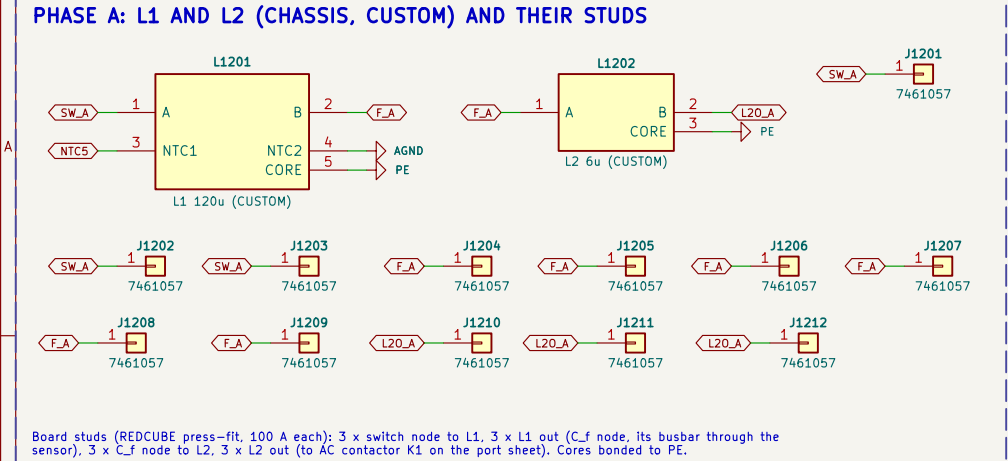
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -0.1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device MOSFET + 2.2k pull the gate to VEE; BG Node -3 vs (1k + FET gates).

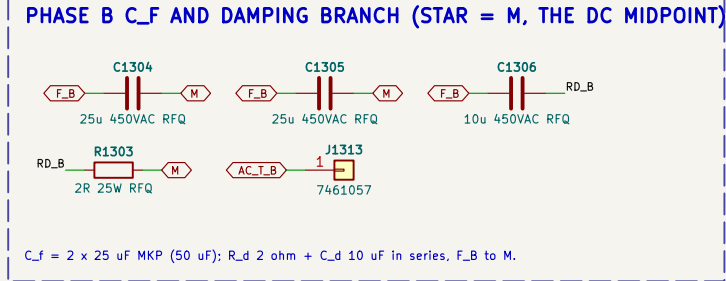
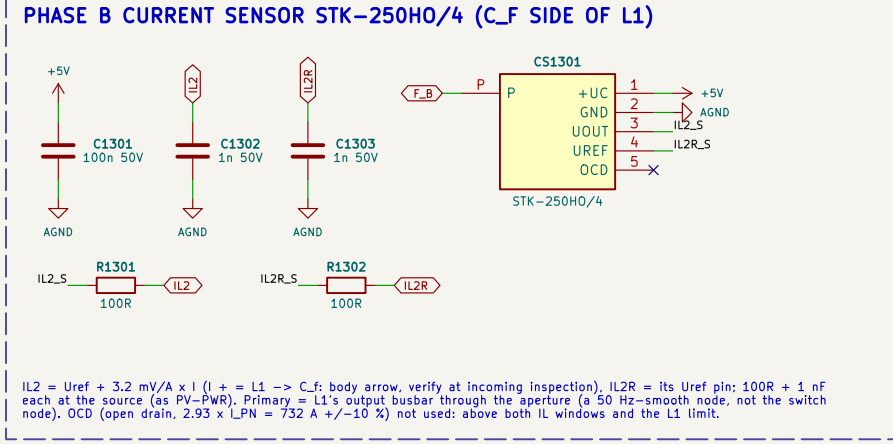
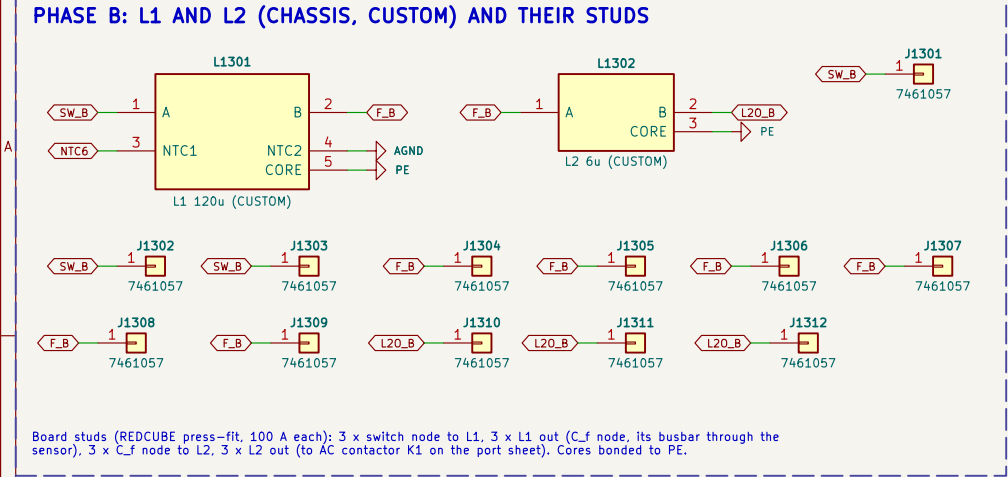
PHC: GATE BIAS (PER PHASE)

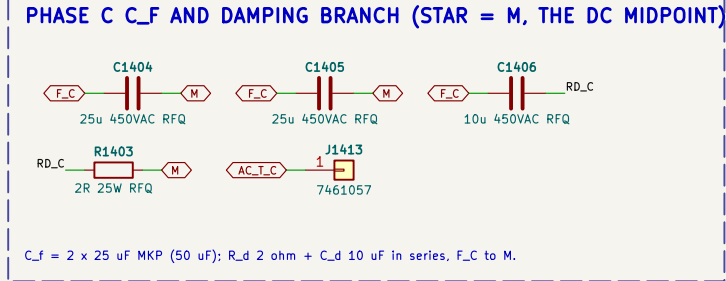
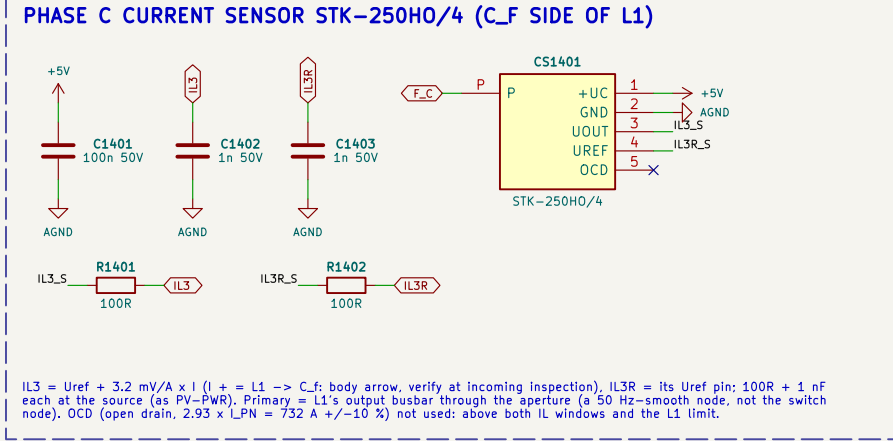
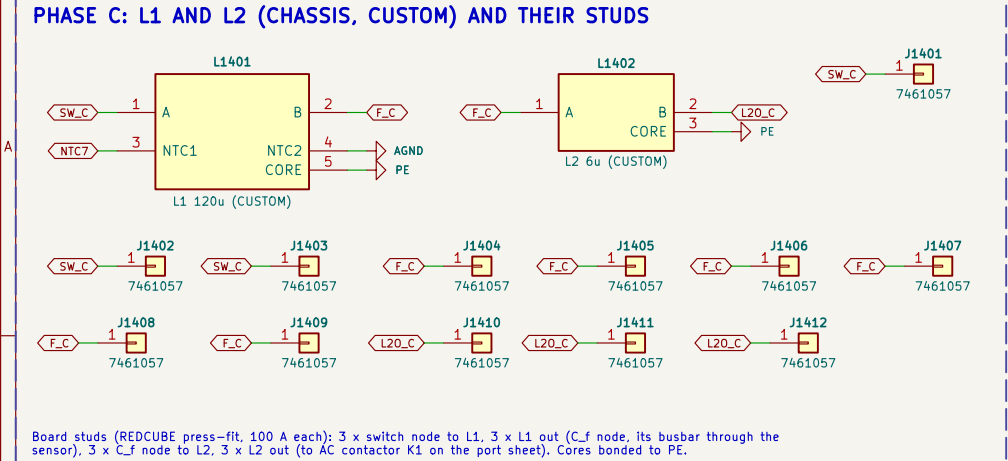


Gate-bias transformer per phase (sim/out/magnetics/design/gate_bias_transformer.json); DMOS: EP17 set (DMR14), ungapped, 4+624 vA, ratio 1:6, barrier 1.77 of high-side winding to primary / 3.43 of adjacent sections, functional insulation PD-free == 1.75 V/kV; 2.18 USD (1 k, ESTIMATE)

PHC050B push-pull 420 kHz from -V_{PH} to V_{PH}; high = on 1.8kV; custom functional insulation
PD-free == 1.75 V/kV, == 5 pF per secondary. Per channel bridge PNP regulatory VDD-VEE; ATLAS1 COM split.



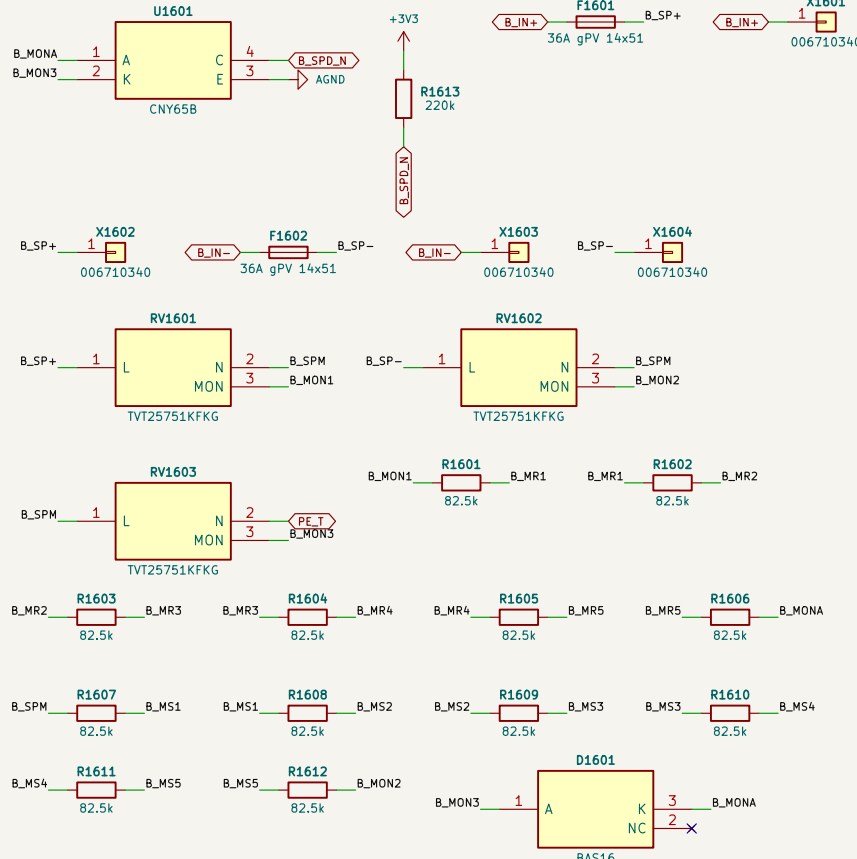




PORT B (BATTERY SOURCE), D-044 LEAN PORT

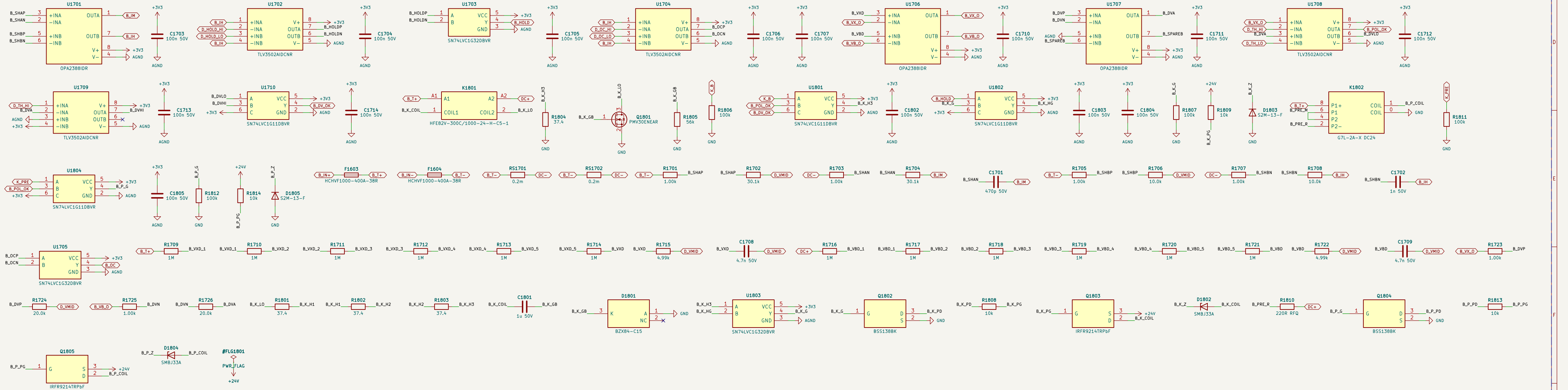
Terminal side BJN+ / BJN-, bank DC+, HCHVF1000-400A-38R aR in both poles.
Contactor opens <= 0.97 kA, held above (port_spec lean). Fuse / contactor coordination: the board's design check.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV25751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up,eff 3793 V pole-PE and pole-pole at 1.5 kA (calc., <= 4.0 V -> reinforced 6 kV / basic 4 kV, D-032); imax 25 kA, Branch fuses ETI CH14x51, gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> SNV658 -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

AR FUSES (BATTERY-TYPE PORT)



HCHVF1000-400A-38R in both poles (aR, partial range): see the board's design check.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 18
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /17 Port B sensing and interlocks/
File: 17_leanB_ctrl.kicad_sch

Title: **PCS-P125 power – Port B sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: A0

KiCad E.D.A. 10.0.6

Id: 18/27

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 19
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /18 Port B coil drives/
File: 18_leanB_drive.kicad_sch

Title: PCS-P125 power – Port B coil drives

Size: A3

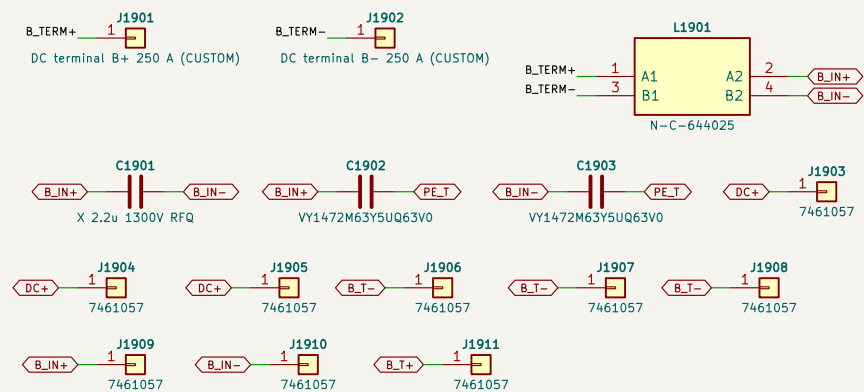
Date: 2026-10-05

Rev: A0

KiCad E.D.A. 10.0.6

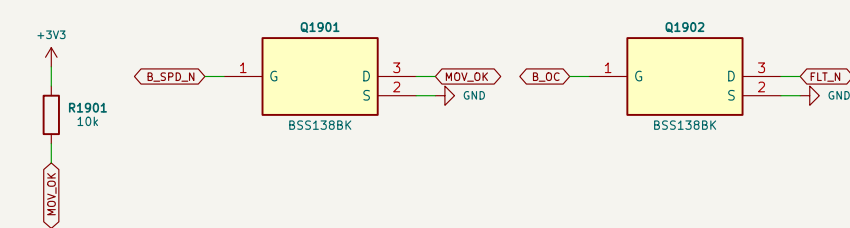
Id: 19/27

DC TERMINALS AND FILTER



B_TERM+/- (250 A), X 2.2 uF 1300 V across the port, Y1 4.7 nF pole - PE_T (removable PE link for the hipot).
Board taps B_IN+ / B_IN- by short links.

PORT STATUS TO PCS_PC



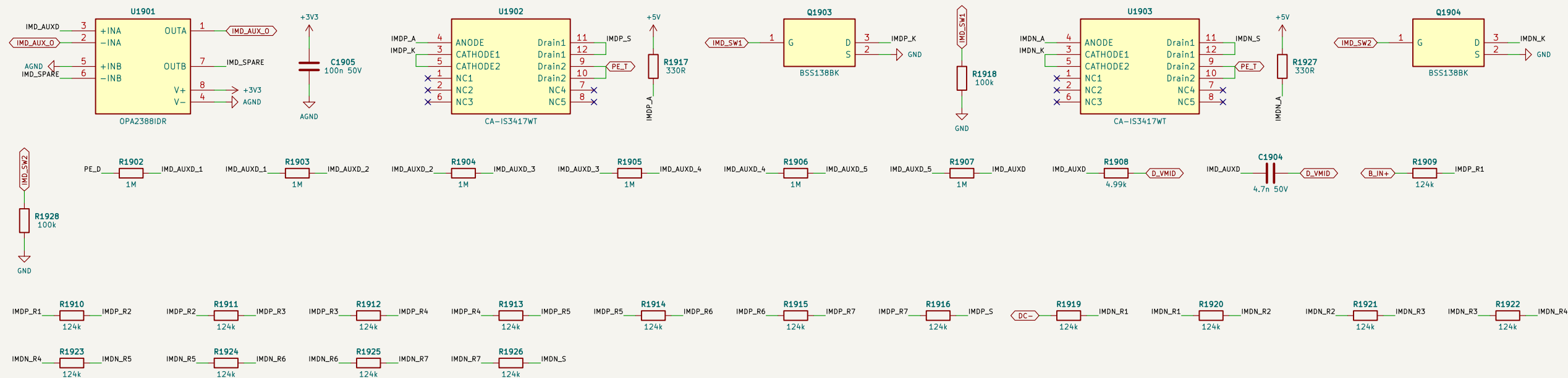
HOLD = B_HOLD (||| >= 0.97-1.03 kA: contactor held). MOV_OK high while the varistor monitor loop conducts. Port over-current (387-413 A) pulls FLT_N low (open drain): the latch trips without firmware.

PE



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot).

INSULATION MONITOR (PORT.LEAN_IMD)



Battery side, contactors open (IEC 62109-2 style check before connecting):
IMD_SW1 = string B_IN+ -> PE, IMD_SW2 = PE -> DC-; VPE to PCS_PC.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 20

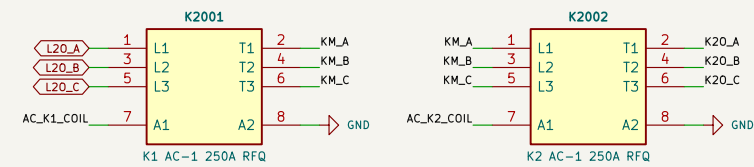
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /19 DC terminals, insulation test/
File: 19_dcterm.kicad_sch

Title: PCS-P125 power – DC terminals, insulation test

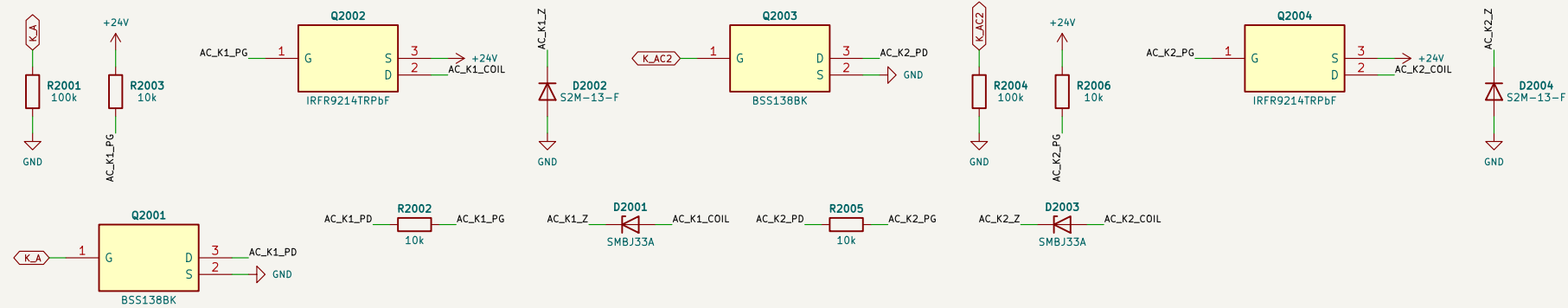
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 20/27

AC CONTACTORS K1, K2 (CHASSIS, RFQ): TWO 3–POLE CONTACTORS IN SERIES



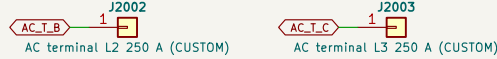
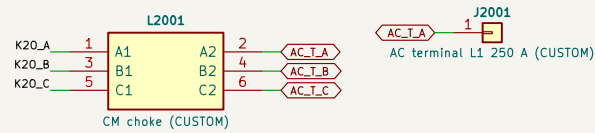
Relay test before every connection: K1 alone, then K2 alone; the voltage across the open one is $V_G - V_C$ (stage-3 dividers). Close only after synchronisation, open only at zero current (firmware).

COIL DRIVERS (PORT_LEAN_DRV): BSS138 → IRFR9214 FROM THE LIVE 24 V, S2M + SMBJ33A CLAMP



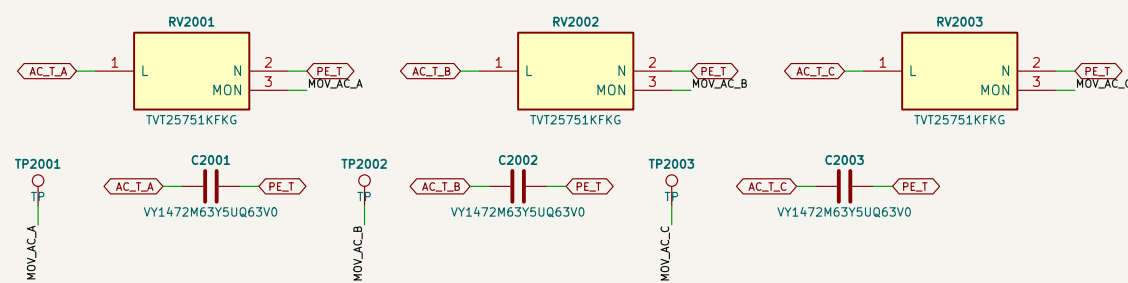
K1 from K_A (= K_AC1), K2 from K_AC2; logic lost = coil off. The contactors' DC coil modules economise.

CM CHoke (CHASSIS, CUSTOM) AND AC TERMINALS



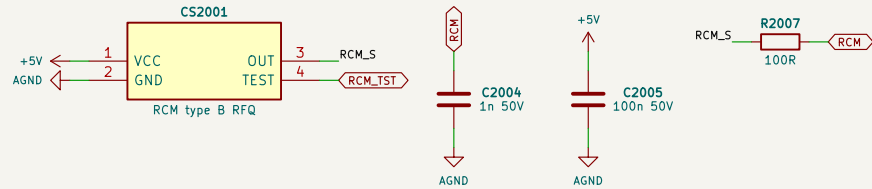
One turn of the three phase busbars through the cores; terminals L1–L3 250 A; PE terminal on the chassis.

AC SURGE PROTECTION (TYPE II) AND Y1



Per phase a thermally protected varistor TVT25751 (465 V AC) and a Y1 4.7 nF to PE_T; the removable PE link opens them for the hipot. Monitor leads to test points (OPEN 5).

RESIDUAL–CURRENT SENSOR (TYPE B, RFQ)



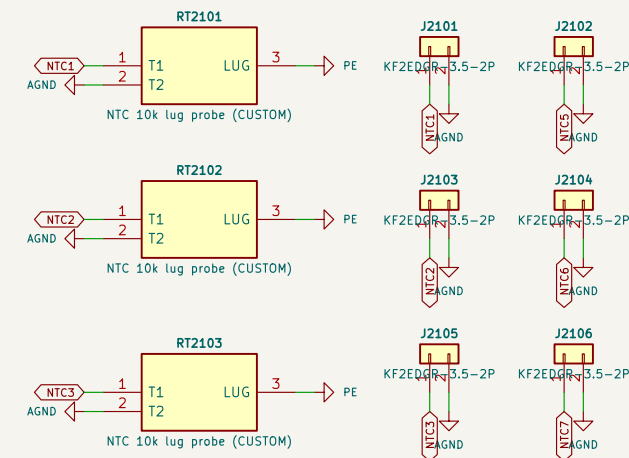
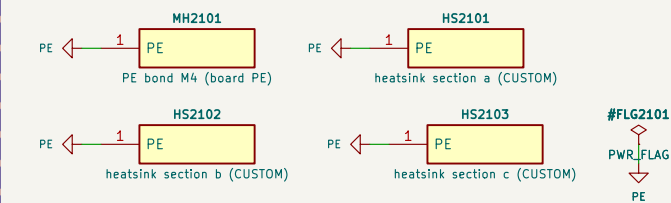
Around L1–L3 between the CM choke and the terminals; RCM analog out and RCM_TST to PCS_X.

Not bench–validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 21
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC–; gate islands functional; HV–PE basic

Sheet: /20 AC port: contactors, SPD, RCM/
File: 20_acport.kicad_sch

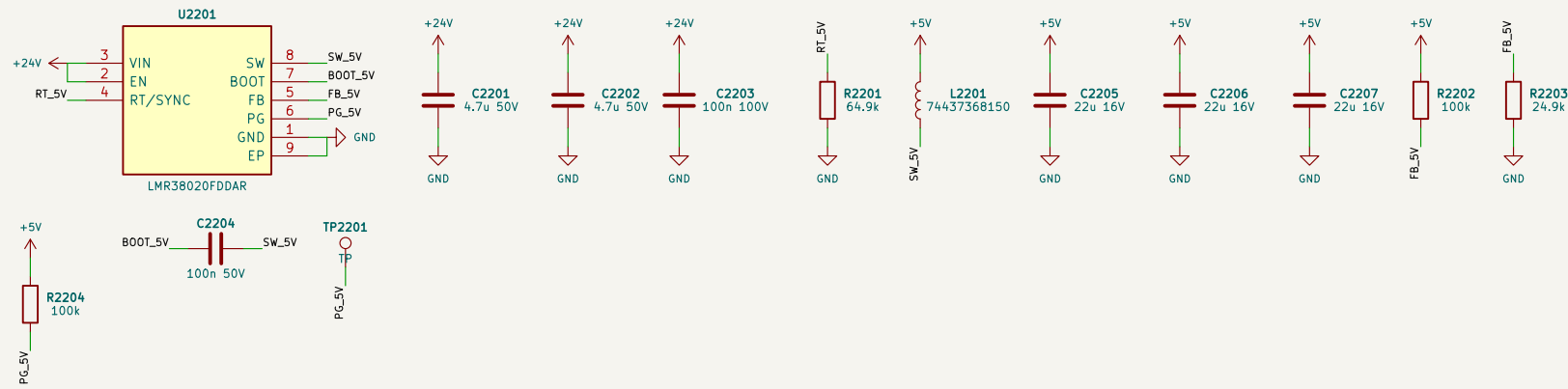
Title: PCS–P125 power – AC port: contactors, SPD, RCM

Size: A3 Date: 2026–10–05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 21/27



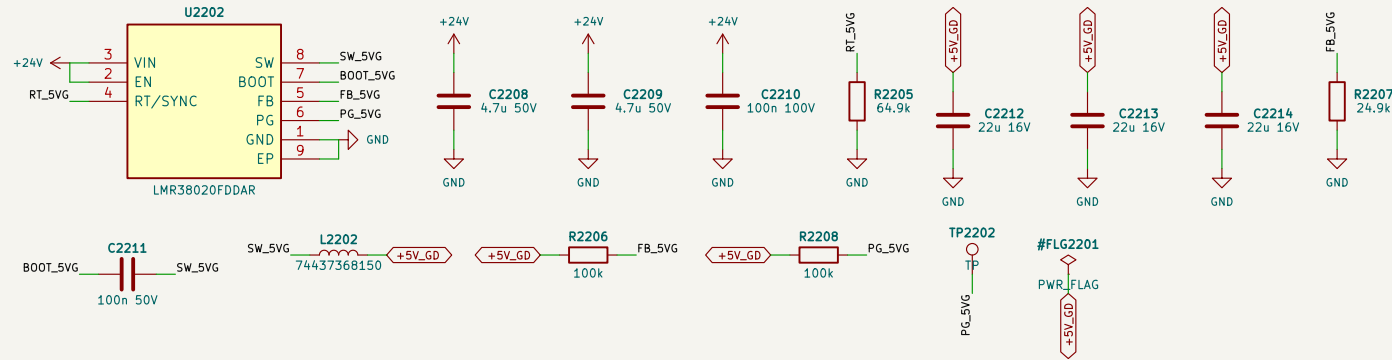
NTC1-3 = heatsink sections a-c: lug probe on the EARTHED heatsink, ≥ 2.5 kV rms lead-to-lug (R-10).
NTC5-7 = L1 winding NTCs (leads on 2-pin headers), 100 nF at each header.

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



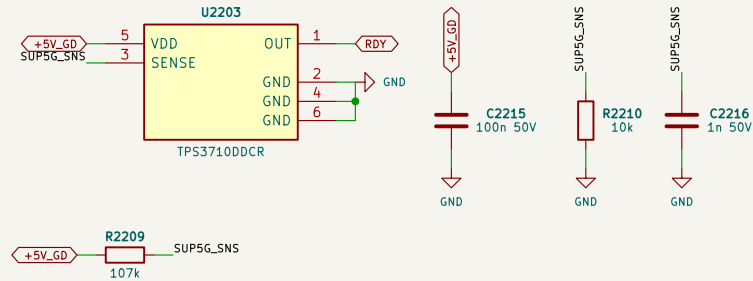
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



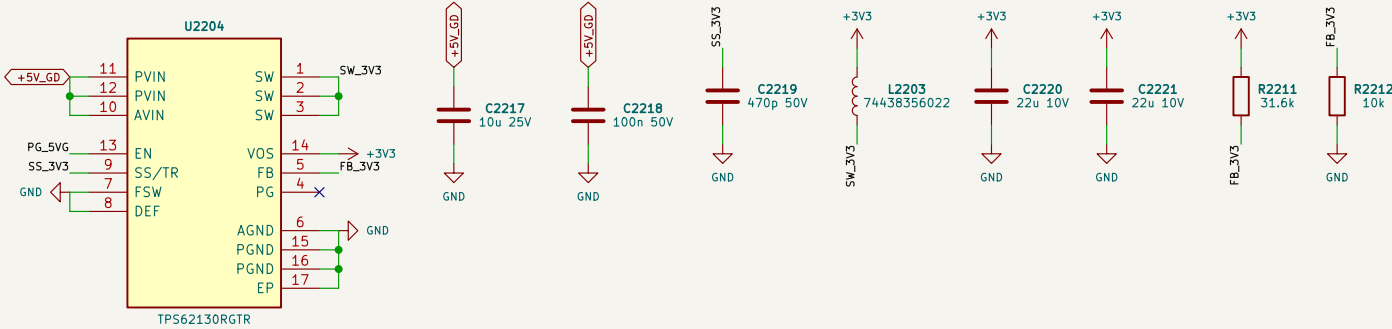
gate bias of legs b and c, +3V3 (TPS62130), ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



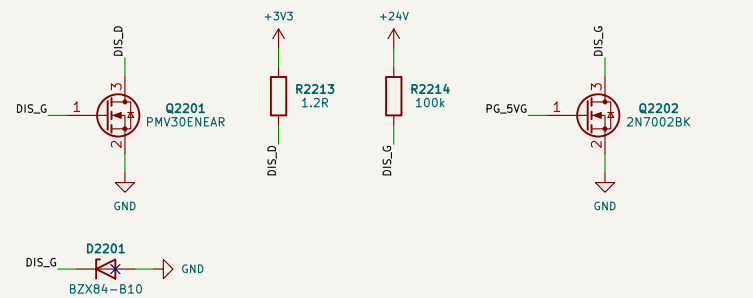
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52-4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5VG, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (>= 3.20 V for the control board's reference), SS/TR 470p COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us), Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3-29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 23
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /22 Live 5 V / 3.3 V, SELV connector/
File: 22_supplies.kicad_sch
Title: PCS-P125 power - Live 5 V / 3.3 V, SELV connector
Size: A3 Date: 2026-10-05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 23/27

TAP B_T+



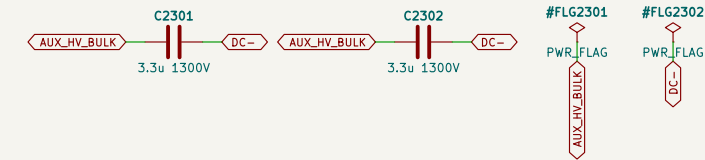
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP DC+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

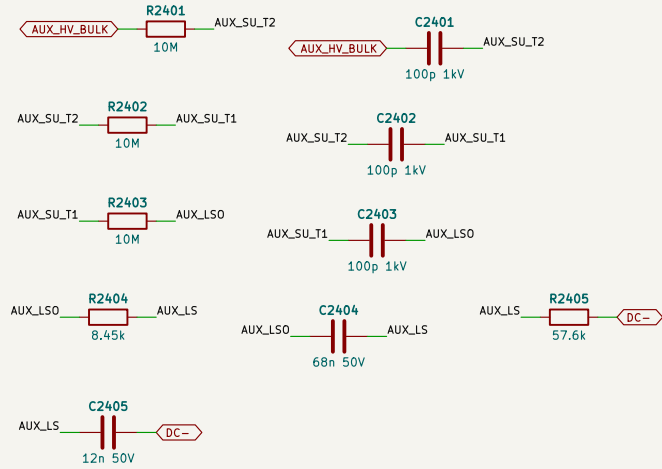
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 24
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /23 AUX75 input taps, OR, film/
File: 23_aux75_input.kicad_sch

Title: PCS-P125 power – AUX75 input taps, OR, film

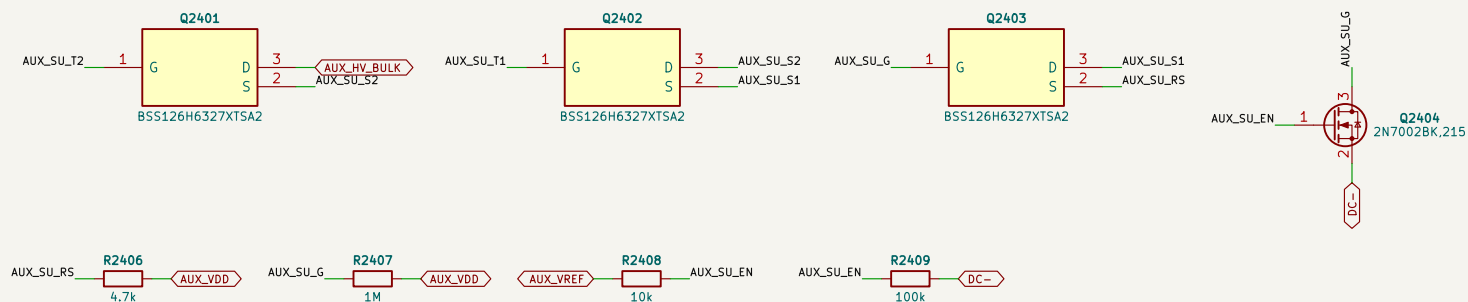
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 24/27

COMPENSATED TAP STRING



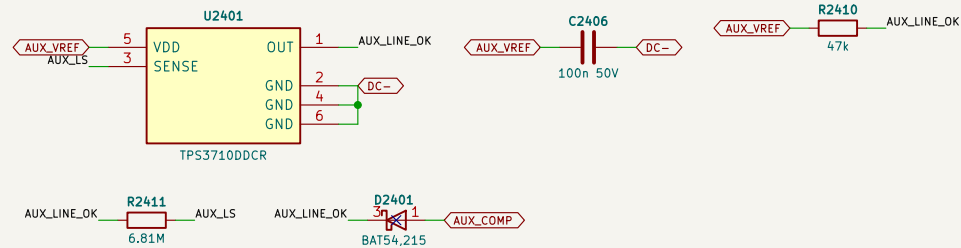
string tap, brown-in 206-215 V, brown-out 179-192 V
brown-in hysteresis (same solve); OV lockout 1107-1162 V

DEPLETION CASCODE CURRENT SOURCE



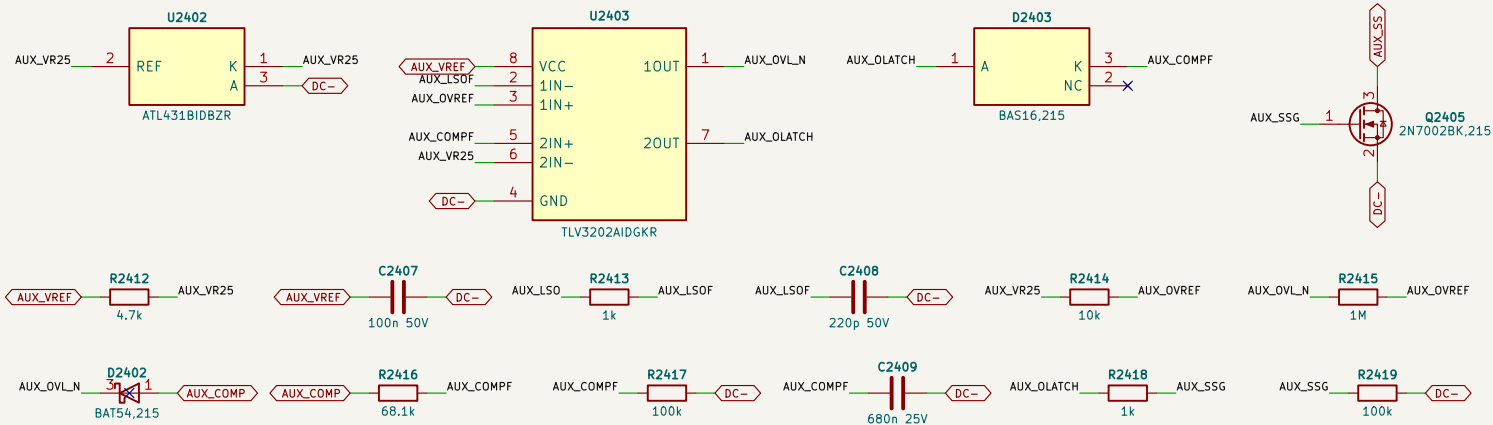
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

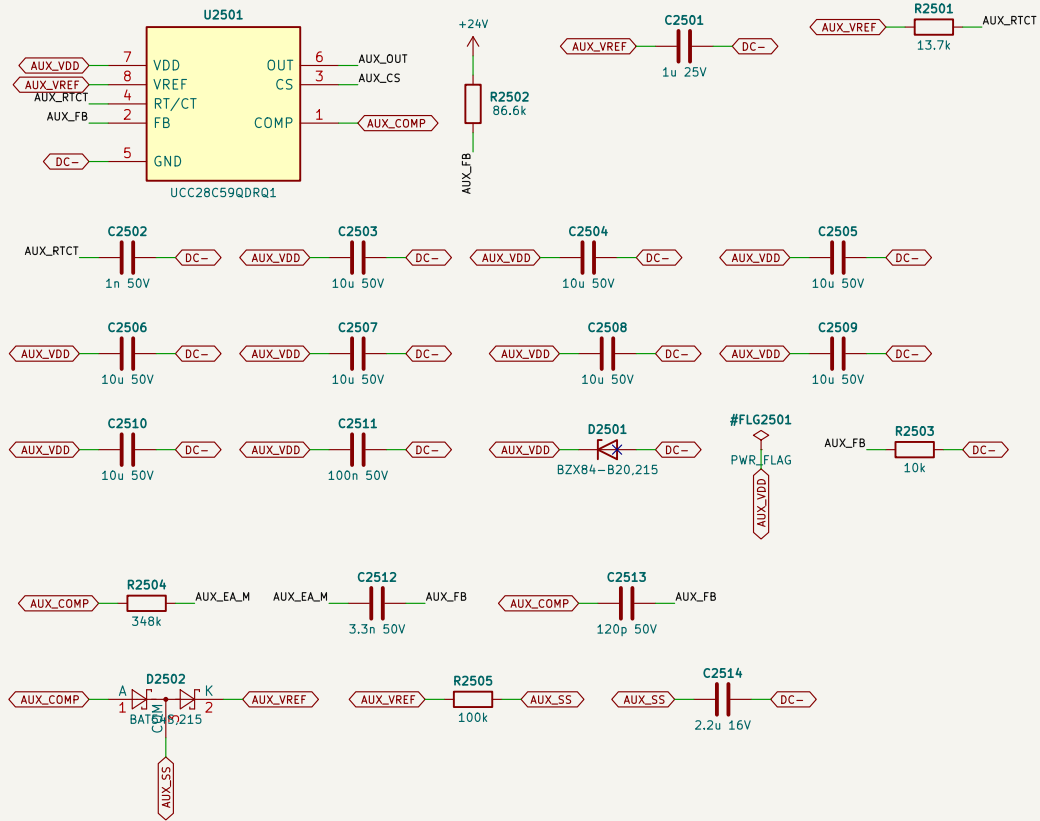
INPUT OV LOCKOUT AND OVERLOAD TIMER



Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35-80 ms -> latch -> SS crowbar -> hiccup.

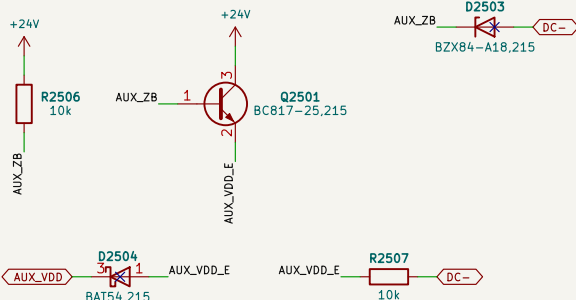
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 25
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

PWM CONTROLLER AND ERROR AMPLIFIER



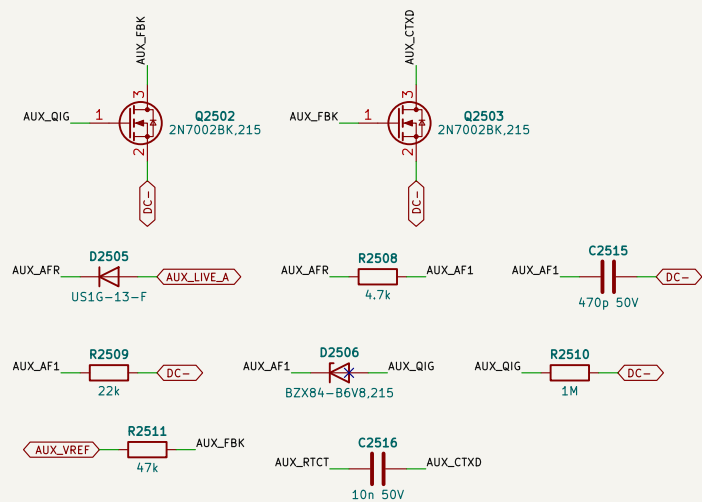
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 241–667 Hz, phase margin >= 58 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



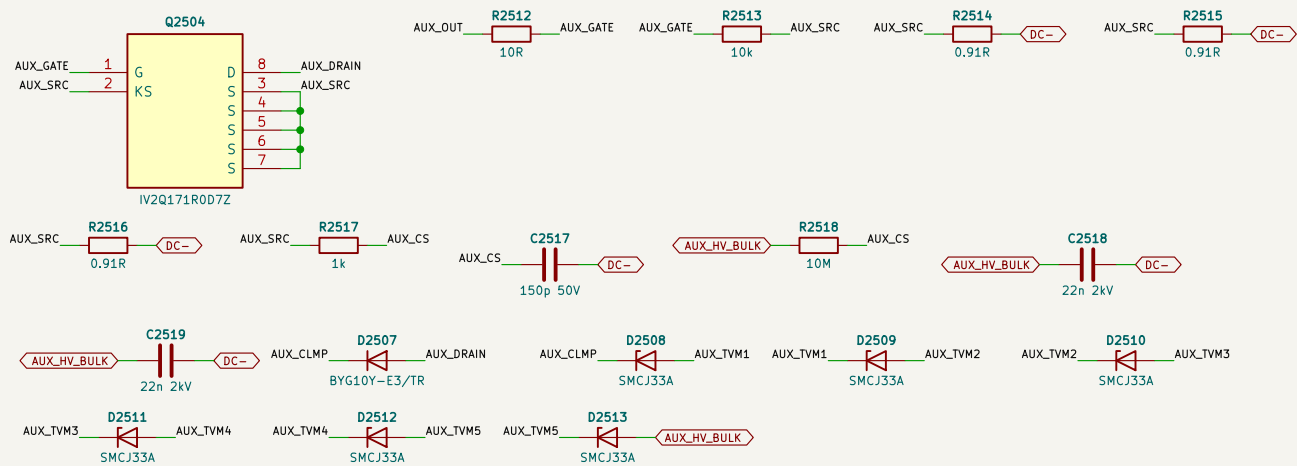
(30 W case, unchanged) follower Zener bias: VDD = gate drive 15.46–18.11 V (BZX84–A18, –40..85 C)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX–03)



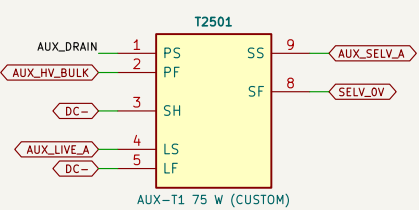
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output -> fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



3 in parallel = 0.303333R: lowest limit 2.90 A delivers 103 W (>= 110 % of 75 W) at every input; highest 3.91 A with the line feed-forward (4.27 A without)
line feed-forward HV_BULK -> CS (CRHV2512): CS offset 101 mV at 1000 V cancels most of the 155–250 ns CS-delay overshoot, so the highest current limit no longer rises with the input
Clamp: BYG10Y + 6 x SMCJ33A; V_DS 1316 / 1420 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE'S REINFORCED SUPPLY BARRIER B1)



Lp 0.485 mH +/-7 %: shield and live winding on DC-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

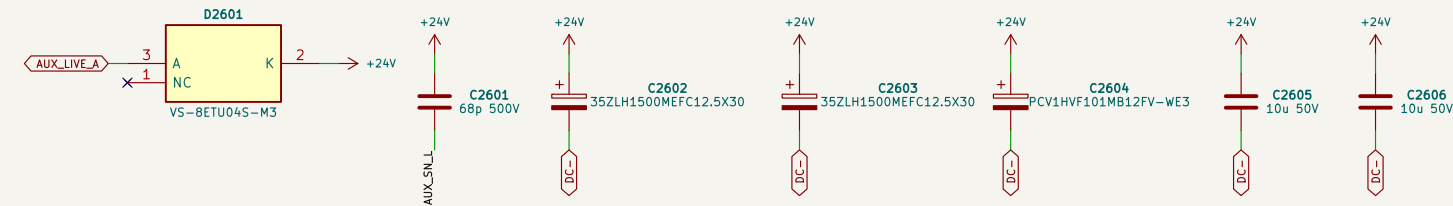
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 26
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /25 AUX75 PWM, SiC, clamp, AUX-T1/
File: 25_aux75_power.kicad_sch

Title: PCS-P125 power – AUX75 PWM, SiC, clamp, AUX-T1

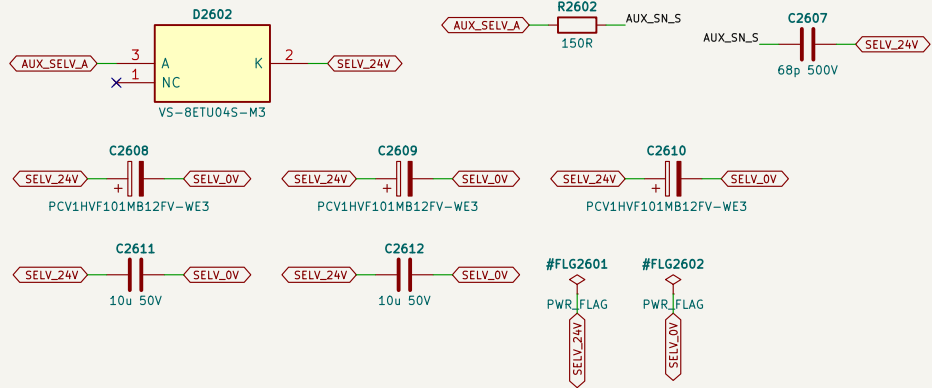
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 26/27

LIVE 24 V RECTIFIER, HOLD-UP



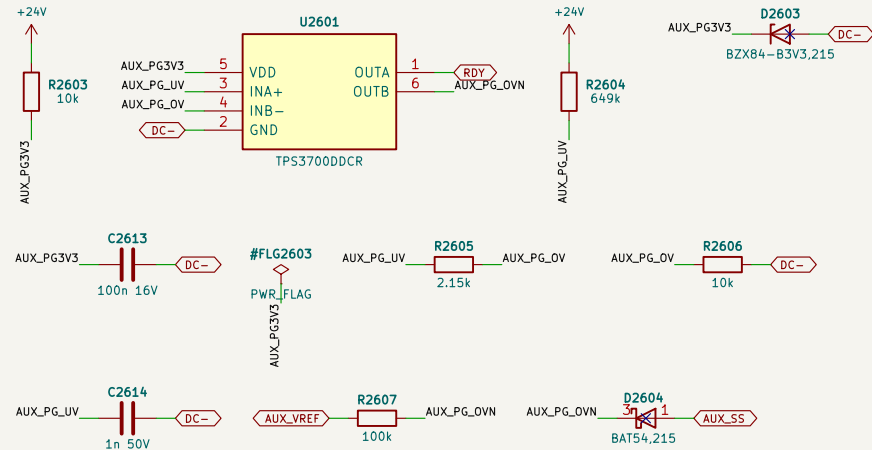
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 27
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /26 AUX75 live 24 V, SELV 24 V, status/
File: 26_aux75_outputs.kicad_sch

Title: PCS-P125 power – AUX75 live 24 V, SELV 24 V, status

Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 27/27